

**When Siblings Drift Apart: Influence of Being Phubbed on Sibling Relationship Quality
and Loneliness among Young Adults**



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A thesis submitted in partial fulfilment of the requirements for the degree of Bachelors of Science in

Clinical Psychology (BSCP) Program

Islamabad, Pakistan

December, 2025

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DECLARATION CERTIFICATE

It is certified that the BS Thesis “When Siblings Drift Apart: Influence of Being Phubbed on Sibling Relationship Quality and Loneliness among Young Adults” written by **Maheen Sohail, Mariya Aman** and **Mutaal Tariq**, with respective Registration No. **BSCP-22S-0204, 0205, and 0206**, has been evaluated by Internal and External examiners.

This thesis has been declared as final.

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AUTHOR'S DECLARATION

We, Maheen Sohail, Mariya Aman, and Mutaal Tariq, state that our BS Thesis titled “When Siblings Drift Apart: Influence of Being Phubbed on Sibling Relationship Quality and Loneliness among Young Adults” is our own work and has not been submitted previously by us for taking any degree from this University (Shifa Tameer-e-Millat) or anywhere else in the country/world.

At any time, if our statement is found to be incorrect, even after our graduation, the university has the right to withdraw our BS degree.

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We solemnly declare that the research work presented in the thesis titled “When Siblings Drift Apart: Influence of Being Phubbed on Sibling Relationship Quality and Loneliness among Young Adults” is solely our research work with no significant contribution from any other person. Any minor assistance/help received has been duly acknowledged, and this complete thesis has been written entirely by us. We understand the zero-tolerance policy of the HEC and the University (Shifa Tameer-e-Millat University) towards plagiarism. Therefore, we, Maheen Sohail, Mariya Aman and Mutaal Tariq, as authors of the above-mentioned thesis, declare that no portion of this thesis has been plagiarized and all material used as reference has been properly cited and acknowledged. We further undertake that if we are found guilty of any form of plagiarism in this thesis, even after the award of our BS degree, HEC and the University shall have the right to publish our names on the HEC/University website, where the list of students who have submitted plagiarized theses is displayed.

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DEDICATION

I dedicate this thesis to my dearest parents, whose unwavering prayers, love, and strength have carried me through every challenge. To my sisters, thank you for believing in me even on the days I doubted myself. And to my friends, your encouragement and thoughtful guidance helped shape this work into what it is today. I am endlessly grateful for each of you.

This work is lovingly dedicated to my parents, whose support and blessings have been the foundation of my entire journey. To my siblings, your quiet confidence in me has been a constant source of motivation. To my friends, my hostel mates and my fellow researchers, thank you for the late-night conversations, shared stress, laughter, and companionship that made the hardest moments bearable. You all played a special role in bringing this thesis to life.

I dedicate this thesis to my father, whose presence in my heart continues to motivate and steady me through every chapter of life. To my mother, thank you for your resilience, warmth, and the prayers that have lifted me at every step. And to my sisters and brother, your unwavering support, small gestures of care, and belief in me have meant more than I can express. I carry this achievement with all of you.

A special note of gratitude goes to our teachers, whose encouragement and unwavering faith in our abilities guided us throughout this journey. Their steady support, thoughtful mentorship, and belief in our potential inspired us to persevere and strive for excellence. We owe much of this accomplishment to their dedication, wisdom, and the support that helped us move forward with confidence.

ACKNOWLEDGMENT

With profound gratitude, we begin by thanking Allah Almighty, the Most Gracious and the Most Compassionate, for blessing us with the perseverance, clarity, and opportunity to carry out and complete this research. Every step of this journey was made possible through His mercy and guidance.

We extend our heartfelt thanks to our respected supervisor, Dr. Syeda Ayat-e-Zainab, whose patient mentorship, meaningful advice, and constant encouragement enriched this study tremendously. Her thoughtful guidance and insightful academic support steered our work in the right direction and strengthened the quality of our research.

We are equally appreciative of the esteemed faculty of the Department of Clinical Psychology, who consistently motivated us and nurtured our academic growth. Their dedication, knowledge, and commitment to our learning have been a source of inspiration throughout this journey.

Our sincere thanks also go to all the individuals who generously participated in our research and facilitated the process of data collection. Their cooperation made this study possible, and we are genuinely grateful for their time and contribution.

We would also like to acknowledge our families and friends, whose prayers, patience, and unwavering support sustained us during challenging moments. Their encouragement has been a constant source of strength.

Finally, we extend our appreciation to everyone who, played a role, directly or indirectly, in helping us complete this work.

Thank you!

Maheen Sohail

Mariya Aman

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ABBREVIATIONS

GSBP	Generic Scale of Being Phubbed
LSRS	Lifespan Sibling Relationship Scale
UCLA- LS	UCLA Loneliness Scale
SPSS	Statistical Package for Social Sciences
APA	American Psychological Association

Abstract

Phubbing, prioritizing phone usage over engaging with the people sitting beside has become devastatingly common within family and sibling interactions. Communication, emotional connection, and the perceived quality of relationships can all be adversely affected by this behaviour. When siblings experience being phubbed, they may feel rejected or less valued, potentially leading to weakened sibling bonds. At the same time, loneliness may emerge as a psychological response to this, suggesting that loneliness could play a mediating role between being phubbed and sibling relationship quality. This study aims to examine the relationship between being phubbed and sibling relationship quality, and to determine whether loneliness mediates this relationship. A cross-sectional design was used with 260 young adults aged 18–30. Data was gathered through purposive sampling. Three different instruments were utilized to assess the study variables i.e. Generic Scale of Being Phubbed (GSBP), Lifespan Sibling Relationship (LSRS), and UCLA Loneliness Scale-Version 3. SPSS 26 was used for analysis. The findings reveal that Being Phubbed by a sibling had a significant moderate negative correlation with overall Sibling Relationship Quality and its subscales. Whereas, significant moderate positive correlation emerged between Being Phubbed and Loneliness. Regression analysis indicated that Being Phubbed by a sibling significantly predicted lower sibling relationship quality, explaining 19.6% of its variance. It also showed that Being Phubbed significantly predicted higher loneliness, explaining 11.2% of its variance. Mediation analysis indicated that loneliness partially mediates the relationship between Being Phubbed and Sibling Relationship Quality. The t-test for independent samples indicated that there is a significant difference between male and female participants. Male participants scored higher on Being Phubbed and Loneliness while female participants scored higher on all dimensions of Sibling Relationship Quality.

Chapter 1

Introduction

The growth and advancement of technology have made it necessary to rely on gadgets like mobile phones when connecting with friends, family, and colleagues. Smartphones have become a key element of global communication. Smartphones offer great convenience, efficiency, and portability, allowing individuals to stay connected, manage tasks, and organize schedules on the go. They enable quick access to emails, information, and mobile payments, making daily activities easier and faster. Additionally, smartphones support education through online research, communication tools, and digital note-taking, while in healthcare, they provide quick and easy access to medical information and applications, leading to productivity and care (Nath & Mukharjee, 2015). In today's age, smartphones have evolved beyond mere communication devices to become essential tools integrated into the day to day lives of people.

Technology and internet play a key role in individuals' daily lives, with smartphones and digital platforms commonly used for learning, social interaction, and entertainment (Parveen & Ramzan, 2024). While these devices can be highly effective tools for staying in touch, excessive use can sometimes create problems in personal and social relationships (Ippolito, 2020). A lot of people are connected to their phones such that it looks like they're addicted (Chotpitayasunondh & Dougals, 2016). Excessive or improper smartphone use has been associated with developing addictive behaviors including dependence, tolerance, and withdrawal (Fischer-Grote et al., 2019).

Smartphone addiction prevalence has been reported around the globe, with it exceeding 60% among adults in United Kingdom (Ofcom, 2021). In India, smartphone addiction was found in 33.33% Women and 46.15% Men among students (Chatterjee & Kar, 2021) and

35.60% among Korean students (Lee & Lee, 2017). The growing evidence connects smartphone addiction to mental health concerns, including anxiety and depression (Guazzini et al., 2019). Studies have demonstrated that smartphone addiction adversely influences the association between self-regulation and well-being in individuals (Mascia et al., 2020).

According to the Pakistan Telecommunication Authority (2020), there were 169 million smartphone users in Pakistan (Rafiq et al., 2020), and as of August 2025, there are 197 million registered mobile devices, as reported by Pakistan Telecommunication Authority (2025). In Pakistan, individuals aged 21–30 make up the largest group of smartphone users, accounting for about 77% and 21% belong to higher age group (Arif et al., 2022).

Recent research revealed that smartphone addiction is a major concern among Pakistani teenagers, impacting nearly 60% of this population (Kamal et al., 2022). In the twin cities of Pakistan, 60% students had smartphone addiction (Khalily et al., 2020). The rapid rise in smartphone usage has also brought negative consequences, especially among young people. Researches indicate that in the higher educational institutes of Pakistan, smartphone addiction negatively relates to academic performance, while students with good time management skills achieve higher grades than those who spend excessive time on smartphones (Khan et al., 2019; Raza et al., 2020). It is reported in Pakistan that actual smartphone addiction relates negatively to wellbeing (Ejaz et al., 2023).

With rising use of smartphones in Pakistan, individuals have difficulty with emotional processing such as alexithymia, a positive relationship has been identified between the two, especially in identifying the feelings (Fatima & Malik, 2024). Interpersonal relationship difficulties and academic stress have also been recognized in association to smartphone addiction (Iqbal et al., 2023). The interpersonal relationship abilities also have been associated negatively with smartphone addiction (Choi & Kim, 2016). Overuse of smartphone contributes

to tension and frustration within the family dynamics which further leads to decrease in face to face communication and emotional disconnection within the family structure (Urieta et al., 2025). Smartphone addicts have poor social interactions as well as unsatisfactory relationship with their families, as the communication gap exists due to smartphone use (Sinsomsack & Kulachai, 2018).

Phubbing

Despite the many benefits smartphones provide, experts in recent years have begun expressing concerns about their potential negative effects on mental and physical health, as well as on social relationships (Khan, 2008). The excessive use of mobile phones has the potential to majorly disrupt daily life, and weaken emotional bonds with close relationships (Elsobeihi & Naser, 2017). This leads to the concept of “Phubbing (Phone Snubbing)”, the act of ignoring other people in a social environment due to preoccupation with one’s smartphone (Chotpitayasunondh & Douglas, 2018). An individual who chooses to focus on their smartphone rather than engage with others during a social interaction is referred to as a “phubber” (Chotpitayasunondh & Dougals, 2016).

A phubber, someone who prioritizes smartphone use over interpersonal interaction, often exhibits poor self-regulation in managing both phone and internet use (Arenz & Schnauber-Stockmann, 2023). This tendency is frequently driven by a compulsive fear of missing out on rewarding experiences and difficulty controlling phone usage (Chotpitayasunondh & Douglas, 2016), reflecting patterns similar to addictive behavior.

Earlier studies have highlighted the harmful effects of phubbing, showing a negative relationship with life satisfaction and a positive association with depression (Parmaksiz , 2021), as well as increased feelings of loneliness and reduced self-esteem (Błachnio & Przepiorka, 2019). Phubbing behavior has been shown to negatively affect interpersonal relationships and

social functioning. Research indicates that excessive smartphone engagement during in-person interactions leads to a reduction in relationship quality and emotional connection (Zhang et al., 2023). Individuals who engage in phubbing often display lower empathy, reduced social competence, and impaired nonverbal communication skills, such as maintaining eye contact (Chotpitayasunondh & Douglas, 2016; Davey et al., 2018). This behavior undermines meaningful communication and contributes to social detachment.

Being Phubbed

The person or people being ignored or dismissed in favor of the phone are known as the "phubbee(s)" (Chotpitayasunondh & Dougals, 2016). Phubbing occurs when an individual's focus shifts from a social interaction to their smartphone, leaving the other person feeling overlooked, a phenomenon termed being phubbed (Al-Saggaf & MacCulloch, 2019). The experience of being phubbed involves feeling ignored or dismissed during communication due to another's phone use (Chotpitayasunondh & Douglas, 2016). This behavior has been classified into contexts such as pphubbing (Roberts & David, 2016) and bphubbing (Roberts & David, 2017).

Phubbing is a passive behaviour, that damages the strength of interpersonal relationship. An individual being phubbed can experience feelings of neglect and elevated vulnerability to mental health disorders. Poor social interaction quality harms phubbees' perceptions of those who phub them (Xie et al., 2020). This leads to negative consequences for communication across every social context, including romantic relationships, parental relationship (Pancani et al., 2021), workplace (Nanda & Prihatsanti, 2023) and other interpersonal and family relationships (Odaci et al., 2024). It negatively impacts relationship satisfaction, emotional detachment and overall sense of personal well-being (Roberts & David, 2016).

Sibling Relationship Quality

The family functions as a central unit of societal stability and holds a vital role in individuals' lives (Crockett et al., 2007). Healthy communication between parents and adolescents is essential for fostering strong relationships and supporting the social and psychological development of young people (Muhwezi et al., 2015).

Similarly, in the family dynamics, siblings play a vital role from the very beginning of life (Cicirelli, 1995). Sibling relationships encompass the entire spectrum of interactions, both verbal and nonverbal, between individuals who share the same parents. They also involve mutual understanding, perceptions, attitudes, beliefs, and emotions toward each other, starting from once a sibling is conscious of the other's existence (Cicirelli, 1991). The early quality of sibling relationships plays a fundamental role in a child's development, as these bonds often represent some of the most stable and persistent ties in a person's life (Lamb & Sutton-Smith, 1982). For those raised alongside one or more siblings, it's clear that these relationships are crucial in shaping the family's social environment and act as a foundation for learning various behaviors. Siblings often serve as example, both positive and negative, of how to behave within the home, and they frequently help one another navigate social experiences beyond the immediate family circle (Stormshak et al., 2009).

Adult sibling relationships is explained in terms of affective, cognitive and behavioural components (Cicirelli, 1991). Sibling relationship, in the young age, helps in learning about emotions and development of emotional understanding (Kramer, 2014). Sibling relationship influences a young adult's understanding of other's emotions and the ability to respond to these expressed emotions (Gungordu & Hernandez-Reif, 2022). Emotional positivity plays a significant role in the quality of adult sibling relationships (Stocker et al., 2020). In the adulthood, sibling relationship further influences the social interactions (Walayat et al., 2025).

Siblings are the salient factors of cognitive development than peers (Azmitia & Hesser, 1993). Having older siblings helps adults in understanding other people's perceptions and feelings (Lo & Mar, 2022).

Healthy sibling bonds have been shown to foster the growth of prosocial behaviors, such as empathy, interpersonal skills, and academic abilities (Patterson, 1984), while also serving as a valuable source of emotional support. For many people, siblings relate as companions, stabilize one another, and support each other meaningfully during times of stress and transition. For some children, relationships with siblings offer meaningful support and contribute to the development of important skills, including emotional awareness and emotional control (Stormshak et al., 2009).

Loneliness

Loneliness is conceptualized as a negative feeling that arises when there is a perceived gap between the level of social interaction an individual desires and the actual quality or quantity of their current interpersonal relationships (Perlman & Peplau, 1982). This interpretation of loneliness brings attention to its emotional dimension, framing it as an unpleasant and distressing state. It also emphasizes on a cognitive aspect, as loneliness results when individuals perceive their current social connections as falling short of their desired standards. Consequently, loneliness acts as a psychological signal pointing to problems or shortcomings in an individual's social connections (Heinrich & Gullone, 2006). It should be noted that not everyone with a limited number of social connections feels lonely (Fischer & Phillips, 1982).

A reciprocal relationship has been identified between teenagers' self-esteem and the quality of their sibling bonds, indicating that supportive and positive sibling interactions can boost adolescents' self-esteem, while higher levels of self-esteem, in turn, contribute to more

positive sibling dynamics (Yeh & Lempers, 2004). If such a bond is not formed between siblings, or it can deteriorate with the fast changing times and technological advancements. Phubbing being socially active virtually, but not present for the relationships around, can lead to conflicts. When a sibling is paying attention to the phone instead to the individual, then the individual is being phubbed. It leads to the perception of social isolation for the phubbee. Experiencing loneliness tends to increase alertness to danger and vulnerability, and also amplifies the need to rebuild social connections (Hawkley & Cacioppo, 2010).

The rise of phubbing behavior is reshaping the way individuals interact within close relationships, including between siblings. As smartphones demand more attention, the quality of sibling interactions may suffer, weakening the emotional bond that is critical for social and emotional development. This weakened connection can contribute to feelings of loneliness, as the emotional closeness typically expected from sibling relationships diminishes. Given these patterns, it becomes important to study the role of being phubbed, sibling relationship quality, and loneliness.

Literature Review

The term “phubbing” was first introduced in an updated edition of the Macquarie Dictionary, created by blending the words *phone* and *snubbing* in May 2012 (Karadağ et al., 2015). Such behavior negatively impacts interpersonal communication, reducing relationship satisfaction and overall quality of life (Roberts & David, 2016). Since phubbing has emerged only recently, existing research on its underlying causes remains limited, and further investigation is needed to understand its contributing factors. Phubbing has been described in varying contexts over the years.

According to Karadağ et al. (2015), “Phubbing can be described as an individual looking at his or her mobile phone during a conversation with other individuals, dealing with the mobile

phone and escaping from interpersonal communication” (p.60). Nazir and Pişkin (2016) described it as “Phubbing is the action of ignoring someone or multiple people during social events and using smartphones, to check or use Facebook, WhatsApp or other social media applications” (p.40). Roberts and David (2017) explained it as “Phubbing is a kind of social exclusion and interpersonal neglect and is used to indicate the interruptions in social relationships caused by mobile phone usage” (p. 207). According to Chotpitayasunondh and Douglas (2018), “Phubbing is the act of snubbing someone in a social setting by looking at your smartphone instead of paying attention” (p.304). Aagaard (2019) discussed it as “Phubbing is the act when one person suddenly turns their gaze downwards and ‘disappears’ into their smartphone in the middle of a social interaction” (p.2). Kaczmarek et al., (2019) delineated “Phubbing refers to adverse behavior that occurs in social situations when individuals maintain their focus on their mobile phones at the expense of a reduced focus on the interlocutor who may feel ignored or snubbed” (p.7). “It is the act of dismissiveness towards someone in a social environment when they look at their phone instead of paying attention to that person.” as explained by Baranova et al. (2022). Salazar (2024) interpreted it as “Technoferences or phubbing refer to the distractions or annoyances usually caused by technological devices when they interrupt a face-to-face interaction.”

Phubbing is a multidimensional behavior emerging from overlapping digital addictions (Karadağ et al., 2015). Mobile phone addiction leads to excessive smartphone use for connection and emotional regulation, often replacing real-life interaction (Turel et al., 2011). Internet addiction involves compulsive online engagement that disrupts daily functioning (Weinstein & Lejoyeux, 2010), while social media addiction reinforces constant online presence and validation seeking through platforms like Facebook and Twitter (Turel & Serenko, 2012). Game addiction further contributes by offering continuous stimulation and escapism (Wood, 2008). Together, these interrelated dependencies illustrate that phubbing is a complex

behavioral pattern that undermines interpersonal communication and fosters social disengagement in the digital age. Findings indicate that the primary predictors of phubbing include mobile phone, SMS, social media, and internet addictions, with mobile phone addiction demonstrating the strongest association (Karadağ et al., 2015).

Research has shown that internet addiction, fear of missing out (FoMO), and self-control are key factors contributing to smartphone addiction, which subsequently influences the extent of phubbing behavior (Chotpitayasunondh & Douglas, 2016). Furthermore, several studies indicate that self-control plays a key role in influencing addictive behaviors, including compulsive smartphone use (Billieux et al., 2007). Individuals with high impulsivity or difficulty regulating their urges may struggle to limit their smartphone use. Thus, smartphone addiction can be viewed as a direct predictor of phubbing, as both involve inappropriate and excessive smartphone engagement (Chotpitayasunondh & Douglas, 2016). People addicted to their devices often find it difficult to refrain from use even in socially inappropriate or restricted settings, demonstrating the behavioral parallels between smartphone addiction and phone snubbing (Billieux et al., 2015; Chotpitayasunondh & Douglas, 2016).

Research indicates that elevated depression, social anxiety, and neuroticism are positively associated with friend phubbing, while agreeableness and conscientiousness show negative links. Increased phubbing also corresponds with lower friendship satisfaction (Erzen et al., 2021; Sun & Samp, 2022).

Moreover, growing evidence suggests that excessive smartphone use extends beyond physical strain, influencing emotional and psychological well-being (Hughes & Burke, 2018). Individuals who engage in prolonged use often experience heightened feelings of loneliness, anxiety, and depressive symptoms, accompanied by increased stress levels and poor sleep

quality (Khan et al., 2023). These findings underscore that the impact of smartphone overuse is not merely technological but deeply human, affecting both body and mind (Song et al., 2025).

Being a phubbee refers to experiencing phubbing behavior from another person (Chotpitayasunondh & Douglas, 2016). Researchers have further contextualized this concept, identifying forms such as “pphubbing,” which occurs when one is phubbed by a romantic partner (Roberts & David, 2016), and “bphubbing,” which describes instances of being ignored by supervisors or boss during interactions (Roberts & David, 2017).

Several theoretical approaches have been developed to describe the dynamics of phubbing and being phubbed. The 2016 conceptual model by McDaniel and Coyne on technology-related interference serves as a foundational framework, illustrating how technology use disrupts interpersonal relationships (McDaniel & Coyne, 2016). Building on this, Roberts and David (2016) introduced the partner phubbing model, linking phubbing behavior to conflicts over phone engagement, which in turn affect relationship wellbeing, life satisfaction, and depression. Their model further incorporated the influence of attachment anxiety as a moderator and clarified the bidirectional relationship between life satisfaction and depression (Roberts & David, 2016). Another model by Benvenuti et al. (2020) examined the associations between self-esteem, self-control, well-being, internet addiction, and phubbing among young adults in Italy. Their findings revealed that phubbing was negatively correlated with self-esteem, self-control, and well-being, while higher internet addiction was related to increased phubbing (Benvenuti et al., 2020).

Research shows quite consistent evidence that a negative relationship exists between partner phubbing and relationship satisfaction (Beukeboom & Pollmann, 2021). Literature indicates that phubbing tends to reduce happiness and emotional closeness within relationships. Over time, this behaviour can weaken intimacy, increase feelings of neglect, and contribute to

overall dissatisfaction between partners (Roberts & David, 2016). When parents engage with their mobile phones during interactions with their children, this behaviour is believed to influence several psychological aspects, particularly by increasing feelings of loneliness and emotional disconnection (Hamami & Widyatno, 2025). When younger family members engage in phubbing, older adults often perceive it as a sign that they are no longer important in their lives or worthy of their time and attention. Phubbing contributes to greater social isolation and emotional distress, alongside a diminished sense of family belonging and overall well-being (Al-Saggaf, 2022).

“Loneliness is the unpleasant experience that occurs when a person's network of social relations is deficient in some important way, either quantitatively or qualitatively” as described by Perlman & Peplau (1981). In Robert Weiss’s typology of loneliness, two different forms of loneliness has been identified. When a person lacks meaningful, intimate relationship with another individual, emotional loneliness stems from it. Individuals who have recently suffered the loss of a close relationship typically experience this form of loneliness. The other type is social loneliness, which unlike emotional loneliness, stems from the absence of a network of social relationships. It is usually experienced in situations when an individual does not belong to a peer group with common interests and shared activities (Russell et al., 1984).

Loneliness differs from social isolation in a sense that it reflects a person’s experience and awareness of not having meaningful social ties. Loneliness is majorly affected by the quality of relationships and the needs that those relationships are expected to fulfill. Feeling unsatisfied with personal connections, such as friendships, romantic bonds, and family ties, significantly contributes to experiencing loneliness (Perlman & Peplau, 1982).

Perlman and Peplau (1982) suggest that feelings of loneliness occur when there is a gap between the level of closeness a person hopes for with others and the closeness they actually

experience. In contrast, other theorists, following the Social Needs Perspective, argue that loneliness results from the failure to satisfy fundamental human social needs (Perlman & Peplau, 1982). While both perspectives focus on the gap between social fulfillment and expectation, the cognitive model emphasizes perception and evaluation, whereas the social needs approach highlights the absence of essential interpersonal bonds.

According to the Jong-Gierveld's model, it views loneliness as a multifaceted experience which stems from the absence of close, intimate bonds, and from a deficiency in broader social connections or group belonging (de Jong-Gierveld, 1987).

Need to Belong Theory discusses that humans are inherently inclined to seek a sense of belonging, which serves as an underlying motivational drive influencing cognitions, feelings, and social conduct. Baumeister & Leary (1995) explores that this need to belong comprises “a pervasive desire to form and maintain at least a minimum quantity of lasting, positive, and significant interpersonal relationships” (p. 497). When individuals struggle to establish or maintain such connections, their unmet desire to belong can result in a sense of loss, often manifesting as loneliness, anxiety, depression, or anger (Baumeister & Leary, 1995; Chipuer, 2001).

The Social Exchange Theory by Blau in 1964 suggests that interpersonal connections operate on a principle for comparing benefits with associated costs. Relationships are maintained when rewards, such as attention, respect, and support, outweigh costs like neglect or rejection. Boss phubbing, when a supervisor ignores an employee to use their phone, disrupts the balance of social exchange. It reduces perceived rewards and increases relational costs, leading employees to feel devalued and excluded. This imbalance can weaken trust, job satisfaction, and overall leader–member relationships (Roberts & David, 2020).

The Attachment Theory by Bowlby in 1969 explains that secure relationships form through consistent care and responsiveness. Parent phubbing, where a parent ignores a child to use their phone, disrupts this emotional availability, reducing the child's sense of security and belonging. Studies show that parental phubbing lowers adolescents' life satisfaction through reduced relationship satisfaction, with stronger effects among preoccupied and fearful teens (Liu et al., 2021). Such neglect fosters loneliness and insecurity (Dong et al., 2023).

Relationship of Being Phubbed and Sibling Relationship Quality

The widespread use of smartphones is contributing to a decline in meaningful social relationships and encouraging more surface-level interactions, especially among younger individuals. This shift is influencing their sense of identity, often leading to increased isolation from family members and a weakening of familial connections (Al-Saggaf, 2022). In the familial setting, problematic smartphone use has been associated negatively with perceived family health (adjusted $\beta = -0.008$), harmony (adjusted $\beta = -0.009$), happiness (adjusted $\beta = -0.015$) and well-being (adjusted $\beta = -0.011$) (Guo et al., 2019).

Being phubbed presents notable challenges to contemporary family life by undermining the quality of relationships and shared interactions. Both parents and children experience a dual impact, while they rely on mobile technology for communication and entertainment, they also express frustration over its disruptive effects on family connection and engagement (Molina et al., 2025). The detrimental effects of overusing smartphones on family harmony and functioning is mostly due to disruption in family communication. The quantity of communication explained the 75% regarding the overall impact of excessive smartphone use on family wellbeing and the quality of communication explained the 94% regarding the overall impact of excessive smartphone use on family wellbeing (Guo et al., 2019).

The bond shared by siblings is typically one of the most persistent, most important, and most positive influence on the emotional, social, and psychological development of individuals throughout the life span (Cicirelli, 1995). Siblings, both in early childhood and adulthood, provide a context through which individuals learn about, love, conflict, cooperation, and empathy. Sibling relationships are also found to be the most powerful, consistent and formative relationship regarding the emotional repertoire and development. The sibling relationship is also an emotionally powerful relationship that is deeply affectionate and intensely ambivalent (Bank & Kahn, 1997). Through shared experiences, both positive and negative, siblings help one another develop emotional understanding and social competence (Dunn & Davies, 2001).

Furthermore, research has shown that supportive sibling relationships can promote emotional competencies including self-regulation and empathy which contributes to psychological well-being (Stormshak et al., 2009). Researchers have shown that the bonds between siblings are among the most complex and powerful relationships influencing individuals' emotional development, social behavior, and overall well-being throughout life in general. As these attachments often resemble other close relationships, they have become important subjects of study in the developmental psychology field (McHale et al., 2012).

When such close siblings bonds are not available due to excessive screen time, it can have detrimental influence on sibling dynamics. Heavy social media use in Pakistan have negative influence on family relationships, as reported by 94% respondents. Individuals feel isolated from family, relatives, friends and even from themselves, which further led to introversion (Ali R. , 2016). Social network sites connect people but it is creating a distance between families due to overuse, it reduces healthy family time, socialization and face to face communication among parents and siblings (Ali et al., 2024) . Close relationships are often taken for granted and it is reported that individuals are more likely to phub those they share close

relationships with than more distant acquaintances (Al-Saggaf & MacCulloch, 2019). This pattern of behavior is altering conventional ways of interacting while negatively impacting people's health and happiness (Al-Saggaf & O'Donnell, 2019). Research addressing phubbing in the context of sibling relationships indicates that heavy use of the phone during in person interactions can undermine sibling emotional connection, communication quality, and perceived relationship satisfaction (Al-Saggaf, 2022). People who share strong, positive bonds with their siblings often experience greater life satisfaction, better social skills, and a stronger sense of emotional resilience. In contrast, when sibling relationships are filled with conflict or emotional distance, individuals tend to struggle more with their emotions and overall well-being (Kim et al., 2006).

Although many studies have focused on phubbing in romantic, parent-child relationships and other interpersonal relationships, very little attention has been given to this behavior and sibling relationship. This lack of research is important to note, as sibling bonds are among the longest and most emotionally meaningful connections people have, playing a key role in shaping their mental well-being, emotional growth, and social development.

Relationship of Being Phubbed and Loneliness

Studies have shown a connection between being phubbed and feelings of loneliness (Ergün et al., 2020; Maftai & Măirean, 2023). Evidence suggests that being phubbed is negatively associated with loneliness and life satisfaction, along with anxiety, negative self, and hostility predicts being phubbed (Ergün et al., 2020). Experience of phubbing makes individuals feel isolated and rejected by their conversation partners, ultimately increasing loneliness (Xu et al., 2022).

Individuals experiencing depression and social anxiety are more likely to be affected by friend phubbing, which in turn lowers the quality and satisfaction of those friendships (Sun &

Samp, 2021). Similarly, when family members engage in phubbing, it can result in heightened feelings of loneliness and depression, while also reducing a sense of family closeness and overall life satisfaction (Al-Saggaf, 2022). Ignoring the presence of your sibling can lead to increase in loneliness. As research indicates that strong affectionate ties between siblings tend to decrease feelings of loneliness (Stocker et al., 2020). Thus, being phubbed by a sibling can foster emotional disconnect and loneliness. Closeness to a sibling is negatively related to loneliness (Ponzetti & James, 1997).

Partner phubbing makes the phubee feel emotionally lonely, which reduces the relationship satisfaction and ultimately minimizes the overall relationship commitment. Being phubbed by a partner (Pphubbing) is a determinant associated with higher emotional loneliness ($b = 0.518, p < 0.001$) and greater emotional loneliness predicts weaker commitment within relationship ($b = -0.562, p < 0.001$), here emotional loneliness partially mediated (Aslanturk & Arslan, 2025). Another research's finding indicated that perceived phubbing was associated with higher levels of loneliness and psychological distress, while loneliness showed an inverse relationship with life satisfaction. It also revealed that loneliness partially mediated the relationship between perceived phubbing and both life satisfaction and psychological distress (Maftai & Măirean, 2023). These results emphasize the impact of perceived phubbing on individuals' mental well-being and underscore the importance of addressing the excessive or inappropriate use of digital devices in interpersonal relationships. Researches concluded there is a notable association between higher levels of loneliness, FoMO, and the tendency to phub (Yaseen et al., 2021).

Gender

Research indicates that phubbing prevalence often varies by gender, though cultural and contextual factors can influence this trend. Studies on gender and phubbing have yielded

inconsistent findings. Overall, most studies report higher rates of phubbing among women (Błachnio et al., 2021). Some studies on university students in Brunei and Mexico, as well as adolescents and adults in Turkey, found women more likely to engage in phubbing behavior (Escalera-Chávez et al., 2020; Balta et al., 2020). However, other research in South America reported different patterns, a study in Ecuador found higher phubbing rates among men (Villafuerte-Garzón & Vera-Perea, 2019), while a Peruvian research found no significant gender differences (Ríos et al., 2021).

A study on sibling relationships and closeness found that men and women generally felt equally close to their siblings. However, the way they expressed or experienced this closeness differed. In simple terms, gender didn't consistently determine whether someone felt "more or less close," but it did shape how that closeness looked or was expressed (Floyd, 1995). In a group of preschool-aged children, girls showed noticeably higher levels of sibling jealousy than boys. However, there were no meaningful gender differences when it came to sibling warmth or conflict (Tsai et al., 2025). The findings show that engaging in a wider range of shared activities was most strongly linked to higher sibling relationship quality in sister–sister pairs. This pattern did not appear in brother–brother pairs, where the association was not statistically significant. This suggests that sisters may benefit more from varied shared activities in strengthening their relationship compared to brothers (Layland et al., 2020).

Gender theory in communication proposes that women generally focus on relational interactions and expressing emotions (Tannen, 1990; Chen et al., 2016), whereas men are more likely to use technology for practical, task-focused purposes (Cai et al., 2017). Higher scores of loneliness are usually reported by men than women, but women acknowledge about being lonely more often because the repercussions for admitting loneliness for men is high (Borys & Perlman, 1985). A more recent meta analysis reported that men and female experience feelings

of loneliness at the same level throughout the lifespan (Maes et al., 2019). Loneliness is more disclosed by younger people than middle-aged, whereas the middle-aged revealed feeling more lonely than older people. People in individualistic cultures described feeling of loneliness more (Barreto et al., 2021).

Theoretical Framework

Need to Belong Theory (Baumeister & Leary, 1995)

The Need to Belong Theory is a fundamental framework in social and interpersonal psychology proposed by Baumeister and Leary in 1995. It proposes that humans possess an inherent tendency to build and maintain long-lasting and rewarding relationships with others. Baumeister & Leary (1995) explores that this need to belong comprises “a pervasive desire to form and maintain at least a minimum quantity of lasting, positive, and significant interpersonal relationships” (p. 497). Humans possess a fundamental need to belong and are highly sensitive to both verbal and nonverbal signals that indicate their social worth (Baumeister & Leary, 1995). Cues suggesting that a person is not sufficiently appreciated or accepted by others tend to reduce self-esteem (Leary, 1999). When individuals perceive cues suggesting that they are undervalued or unwelcome, such as being ignored or excluded, they tend to experience feelings of rejection and various negative psychological outcomes, including a threatened sense of belonging and loneliness (Wesselmann et al., 2016). Unfortunately, such experiences of social exclusion are relatively common in everyday life (Nezlek et al., 2012).

Social exclusion can take many different forms, ranging from overt rejection to subtle acts of neglect. At times, individuals may be directly excluded through explicit remarks that communicate they are not welcome within a group (Twenge et al., 2001). Even nonverbal behaviors, such as mocking or derisive laughter, can foster a sense of rejection and emotional discomfort (Klages & Wirth, 2014). However, exclusion often occurs in more indirect and

understated ways, where a person is simply ignored rather than openly rejected. This can manifest through avoiding acknowledgment of someone's presence, commonly referred to as the silent treatment or cold shoulder (Williams, 2002). In digital contexts, similar behaviors appear in the form of "ghosting", when individuals deliberately ignore calls, messages, or social media interactions (Freedman et al., 2019; Timmermans et al., 2021). Overall, whether through deliberate words or passive inattention, these actions can deeply undermine a person's feeling of inclusion and therefore result in loneliness.

The subtle nonverbal cues, such as avoiding eye contact or prolonged silence, can lead to feelings of exclusion. Participants who were denied eye contact reported lower belongingness and self-esteem, and pedestrians who were "stared through" felt more socially disconnected than those who received eye contact (Wirth et al., 2010). Similarly, uncomfortable silences in interactions have been found to diminish individuals' sense of inclusion and social worth (Koudenburg et al., 2011).

Individuals may inadvertently create a sense of neglect or exclusion by engaging with their phones during in-person interactions. When someone prioritizes their device over the person they are with, referred to as phubbing, it can result in diminished feelings of connection, satisfaction in the relationship and exclusion (Hales et al., 2018).

Phubbing, using a phone during in-person or romantic interactions, creates exclusionary cues, such as disrupted eye contact or conversational breaks. Since gaze direction plays a key role in fostering social connection, phone-focused attention can signal disinterest and hinder feelings of affiliation during interactions (Vanden Abeele & Postma-Nilsenova, 2018). When partners spend time together, they often anticipate full attention from one another; however, when this expectation is unmet, it can lead to negative emotions toward both the partner and the interaction (Miller-Ott & Kelly, 2015).

When individuals experience social exclusion, such as being phubbed, they undergo a process similar to that described by Williams (2009). The phubee initially experiences loneliness, emotional distress, and threats to belonging and self-esteem. In line with the Need to Belong Theory, this exclusion triggers efforts to restore connection by seeking new relationships or repairing existing ones. If reconnection fails, individuals may withdraw socially, and prolonged unmet belonging needs can result in reduced relationship satisfaction, frustration, or antisocial behavior. Being phubbed over a trivial reason can feel hurtful, whereas when it occurs for a meaningful or justified reason, the emotional impact tends to lessen or even disappear (Wesselmann et al., 2016; Baumeister & Leary, 1995).

Cognitive Discrepancy Model

The focus is primarily on the meaningfulness of social bonds rather than on quantity of social relationships. A person may be surrounded by a bunch of people and still end up feeling lonely if he is constantly being phubbed by his companions. Being ignored or not given desired attention in co-present interactions may give the impression of lack of meaningful social relationships and dissatisfaction in relationship along with negative emotions (Amichai-Hamburger & Etgar, 2016; Wang et al., 2017). This gives rise to a gap between a person's expected relationship quality and the actual quality of relationship that the person subjectively experiences. This phenomenon was conceptualized by Perlman and Peplau (1982) in their cognitive discrepancy model.

Perlman and Peplau (1982) define it as “a more cognitive perspective, emphasizing the match between a person's desires or expectations for relationships and the reality of his or her social life. This is known as the cognitive discrepancy model of loneliness” (p. 572). The desired social relationships represent the kind and degree of social connection a person ideally seeks. This includes both the quantity of relationships, how many meaningful connections they

wish to have and the quality of those relationships, such as their depth, support, and emotional closeness. Essentially, it reflects an individual's personal expectations or standards for social connection (Perlman & Peplau, 1982).

According to this model, feelings of isolation emerge when there is a discrepancy between hoped-for and real-life social interactions. In other words, when the quality or quantity of social connections a person wants does not match what they currently experience, loneliness arises. Cognitive factors such as how individuals interpret their social situation, compare themselves to others, or attribute blame for their loneliness can intensify these feelings. The model emphasizes that loneliness is not just about being alone but about perceiving a disconnect between what one expects from social relationships and what one actually receives (Perlman & Peplau, 1982).

This framework can be applied to the experience of phubbing within sibling relationships. When siblings engage in phubbing, diverting their attention to smartphones during shared moments, it violates the expectation of mutual attention, warmth, and engagement that typically characterizes sibling bonds. Being phubbed by a sibling may therefore create a discrepancy between the desired closeness and the perceived reality of the relationship, where one's need for closeness and connection is unmet. Over time, repeated experiences of being ignored or dismissed by a sibling can heighten feelings of loneliness and relationship dissatisfaction, as individuals cognitively interpret these behaviors as signs of devaluation or rejection.

Rationale

In this digital age, smartphones have become integral to daily life. Approximately 98% of Internet users own at least one smartphone and spend over four hours online using it each day (Young, 2017). Recent data collected by GSMA Intelligence in 2025, indicates that around 6.9 billion individuals globally are smartphone users and 55% of these users are from Asia. In the context of Pakistan, smartphone usage has grown remarkably over the past decade from just 10% in 2014 to approximately 51% by 2020 (Schumann et al., 2022). Among these users, around 77% are between the ages of 21 and 30, reflecting a young and digitally active population (Arif et al., 2022).

Smartphones have transformed the way we communicate, but they have also disrupted many face-to-face interactions. The pervasive use of smartphones has given rise to the phenomenon of "phubbing", the act of ignoring someone in a social setting by focusing on one's phone instead. It has become increasingly common in everyday interactions for individuals to ignore those around them due to smartphone use (Chotpitayasunondh & Douglas, 2018). People can often be both the ones who phub and the ones who get phubbed. Ignoring someone during a conversation because of a phone can lead the other person to do the same in return. (Chotpitayasunondh & Douglas, 2016).

The focus of this study is on phubbing in the context of sibling relationships. Sibling relationships hold a deep emotional value, especially in Asia, where families are closely knit and rooted in culture. In many households, siblings aren't just people a person grow up with, they're lifelong companions, trusted confidants and emotional partners. Sibling relationships, built on shared experiences and emotional closeness, play an important role in shaping individual growth and well-being. Sibling relationships are often the longest-lasting interpersonal bonds in a person's life and are critical to one's emotional and psychological

development (Buhrmester, 1992). They play a foundational role in identity formation and serve as a consistent source of social and emotional support throughout various life stages. Particularly during adulthood, sibling relationships become more emotionally complex and evolve into friendship-like bonds, characterized by mutual reliance, empathy, and shared experiences (Bank & Kahn, 1997). Sibling relationships provide emotional support and help shape a person's identity. In many Asian cultures, siblings often act as even second parents, guiding daily life and important decisions. These close bonds strengthen emotional well-being, social growth, and family harmony. Despite their importance, research on the impact of digital distractions on these relationships remains limited.

It is also important to consider the Islamic perspective on sibling relationships and mobile use especially in a Muslim-majority country like Pakistan. Islam teaches the importance of balance in all aspects of life, including the use of technology and smartphones, encouraging people not to spend too much time on any single activity (Karimulloh & Rahayu, 2024). The religion also emphasises strong and caring relationships between relatives including siblings and views the family as a source of love, mercy, and support. The MuslimFamilyHub (2023) highlights that strong family ties are encouraged in the Qur'an and Hadith, emphasising that siblings and other relatives should treat each other with kindness, compassion and care. Islam places great importance on the well-being of children, treating siblings fairly, and maintaining close relationships with relatives. These are considered essential aspects of living a faithful and responsible life and the Quranic teachings warn us that cutting ties with family members is a serious matter (Zarabozo, 2007). When digital habits like phubbing troubles these important family values, it is important to consider the impact of modern technology on the special bond between siblings. Ultimately, Islam encourages strong bonds between siblings, showing that caring and attentive relationships are essential for a happy and balanced life.

Research indicates that phubbing, or giving attention to a smartphone instead of the people around you, is becoming more widespread as smartphone use is increasing. A review of more than 60 studies highlighted the causes and consequences of phubbing across different social settings, showing that it can weaken interpersonal connections (Shahzadi et al., 2024). Studies highlight that individuals who are “phubbed” by others such as family members or partners often report lower levels of relationship quality and greater feelings of isolation (Xie et al., 2022). Research shows that strong sibling relationships contribute to social development, self-esteem, and help individuals cope with challenges in life (Dunn, 2007). Loneliness refers to the subjective feeling of being socially isolated or lacking meaningful connections. Research shows that older adults who experience conflicted sibling relationships report higher levels of loneliness, while warmth and supportive sibling bonds are linked with lower loneliness and better overall well-being (Stocker et al., 2020).

The phenomenon of being phubbed has been deeply studied in the context of romantic and parent-child relationships. Studies have demonstrated that phubbing in romantic partnerships can lead to decreased relationship satisfaction and increased feelings of loneliness and psychological distress (Roberts & David, 2016). Similarly, research on parental phubbing has linked this behavior to weakened parent-child bonds, reduced emotional connection, and adverse effects on children's social-emotional development (Hong et al., 2019). This study aims to examine the role of digital distractions in shaping the quality of sibling relationships in everyday life considering the importance and emotional support these relationships provide. So far, limited research has examined the role of phubbing in family relationships, including those with parents, children, and siblings (Al-Saggaf & O'Donnell, 2019).

This study aims to investigate the relationship between being phubbed and sibling relationship. If being phubbed leads to increased feelings of loneliness and deteriorates the

quality of sibling interactions, it could have long-term implications on individual well-being and family dynamics. By addressing this aspect, the research aims to fill an important gap in the literature and offer a clearer understanding of the relationship between digital behaviors and sibling bonds.

This research may help us understand phone-based distractions among siblings, such as phubbing, can influence emotional closeness, loneliness and overall relationship quality, offering important implications for family functioning, relational dynamics and personality-related tendencies. In many families today, smartphones often compete with face-to-face interaction for attention even small, repeated instances of being ignored in favour of a phone can gradually weaken emotional bonds and family cohesion. Prior research supports this, showing that parental phubbing is associated with greater adolescent loneliness and perceived declines in family closeness (Hamami & Widyatno, 2025). Similarly, another research highlights that phone-based distractions reduce emotional availability and shared meaning in family interactions (Shahzadi et al., 2024). Moreover, studies indicate that supportive sibling relationships contribute significantly to overall family well-being, suggesting that disruptions caused by phubbing could have broader consequences for familial harmony (Milevsky, 2011). Investigating phubbing among siblings can reveal the impact of digital distractions on family bonds, values and connection.

Every person brings their own unique personality to a relationship, and these differences can influence the way they reach for their phones around each other and their experience of being ignored. For example, research among Generation Z students found that those who were more conscientious or extraverted tended to engage in less phubbing, while higher levels of neuroticism were linked to more frequent phone-based distractions (Ashrafifard et al., 2024). Other studies have shown that FoMO is associated with increased phubbing behaviors and

personality traits such as boredom proneness and low openness have also been connected to higher levels of phubbing (Ansari et al., 2024). Looking at these individual differences in the sibling context can help explain why some siblings are more inclined to use their phones during shared time or feel more disturbed when they are phubbed, providing insight into the ways personality shapes both digital habits and the quality of sibling interactions.

Healthy relationships survive on attention, empathy, and emotional responsiveness. When one frequently turns to their phone, these qualities can be disrupted, leaving the other feeling neglected and less satisfied with the relationship. Research in interpersonal relationships has shown that being phubbed is linked to lower relationship satisfaction, with loneliness playing a key mediating role (Xie et al., 2022). Studies among peers similarly indicate that frequent phubbing can reduce emotional closeness and a sense of connection (Büyükşakar & Celik, 2024). Applying these findings to sibling relationships highlights the importance of presence and attentive engagement, suggesting that minimizing phone-based distractions may be crucial for nurturing strong, supportive bonds.

This research may therefore shed light on the processes that help siblings maintain emotional closeness in today's digitally connected world. The findings are expected to highlight guidelines that promote meaningful interactions among siblings, offering a clearer understanding of the role phubbing plays in family relationships and communication particularly in the context of sibling bonds.

In this study, the focus is specifically on individuals aged 18 and above, both for participants and the siblings they report on. This age group was chosen because adult siblings are more emotionally developed and better equipped with the social awareness needed to interpret, reciprocate, and respond to emotionally charged interactions like phubbing. In contrast, younger siblings under 18 are still developing key skills like empathy, perspective-

taking, and social understanding (Eisenberg et al., 2006), which may make them less sensitive to or aware of such behavioral nuances. It was found that loneliness was higher in adolescents and young adults during university studies compared to those who are in high school (Jaud et al., 2023). Moreover, in our study, participants who have multiple siblings are instructed to focus on the sibling they are most emotionally closest to and co-resident with. The participant is more likely to provide accurate and relevant information about this sibling due to their closer relationship and shared living situation. This approach helps ensure that the data collected is more consistent and reliable, making the analysis easier and more meaningful.

Objectives

1. To examine the relationship between being phubbed by a sibling and sibling relationship quality among young adults.
2. To investigate the influence of being phubbed on affective, behavioral, and cognitive aspects of sibling relationship quality and loneliness among young adults.
3. To study the mediating role of loneliness between being phubbed, sibling relationship quality and its affective, behavioural and cognitive dimensions, among young adults.
4. To study gender and age based differences between being phubbed, sibling relationship quality and loneliness among young adults.

Hypotheses

- H1. Being phubbed by a sibling is negatively associated with sibling relationship quality and its affective, behavioral, and cognitive dimensions among young adults.
- H2. There is a positive relationship between being phubbed and loneliness among young adults.

H3. Loneliness mediates the relationship between being phubbed, sibling relationship quality and its affective, behavioural and cognitive dimensions, among young adults.

H4. There are significant gender based differences between being phubbed, sibling relationship quality, its affective, behavioural and cognitive dimensions and loneliness among young adults.

Conceptual Framework

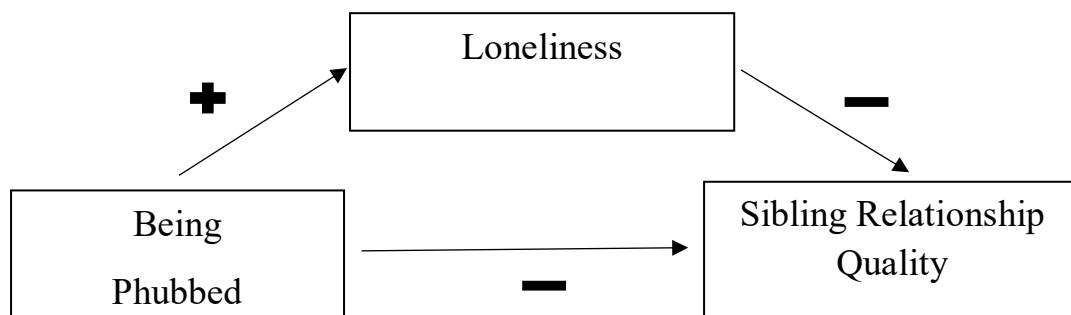


Figure 1. Conceptual Framework of the Study

Chapter 2

Methodology

Research Design

The primary goal of this study is to investigate the relationship between Being Phubbed, Sibling Relationship Quality, and Loneliness and to study the role of Loneliness as mediator between Being Phubbed and Sibling Relationship Quality among young adults. This study employed a Quantitative Cross-Sectional Research Design.

Sample

The sample was taken using purposive sampling method. Data was gathered from around 260 participants who have at least one sibling older than the age of 18 with whom they are currently living.

Inclusion Criteria

- The individual must be above the age of 18 and have at least one sibling above the age of 18
- The sibling must have a habit of daily phone usage
- The individual must be co-residing with the said sibling

Exclusion Criteria

- Individuals with siblings who have mental health problem or have substance use disorder

Operational Definitions

Being Phubbed

The term “phubbing” represents the act of snubbing someone in a social setting by concentrating on one’s phone instead of talking to the person directly. Being a phubbee involves being at the receiving end of phubbing behavior (Chotpitayasunondh & Douglas, 2016).

It was operationalized using the **Generic Scale of Being Phubbed** (Chotpitayasunondh & Douglas, 2018), with items adapted to reflect sibling-specific behaviors (e.g., “My sibling checks their phone while we are talking”). Higher scores on GSBP indicate that the individual has more frequent experiences of being phubbed (Büber et al., 2025).

Loneliness

Loneliness has been defined as the aversive state experienced when a discrepancy exists between the interpersonal relationships one wishes to have, and those that one perceives they currently have (Perlman & Peplau, 1982).

It was operationalized using the **UCLA Loneliness Scale – Version 3** (Russell, 1996), which captures perceived disconnection and dissatisfaction with one’s social relationships. Higher scores on UCLA Loneliness Scale indicate greater loneliness (Shaygan et al, 2025).

Sibling Relationship Quality

The term "Sibling Connection" refers to the total of the interactions (actions, verbal and nonverbal communication) of two (or more) individuals who share common parents, as well as their knowledge, perceptions, attitudes, beliefs, and feelings regarding each other from the time when one sibling first became aware of the other. The term "sibling connection" is used interchangeably with the term "sibling relationship"; it is regarded as having behavioral, cognitive, and affective components and as existing over an extended time period (Cicirelli, 1991).

It was operationalized using the **Lifespan Sibling Relationship Scale** (Riggio, 2000) focusing on subscales relevant to adult affect, behavior, and cognition. Higher scores indicate a more positive sibling relationship quality while lower scores indicate a less positive sibling relationship quality (Lin et al. 2024).

Instruments

Informed Consent

The participants were given a consent form attached at beginning of the research questionnaire. It contained all the important information related to the study, aim of the study, who is undertaking it, total time taken to fill the questionnaire. It also outlined rights of the participant for voluntary participant, anonymity of participant, limits of confidentiality, any possible risks, and the right to withdraw at any point without consequences.

Demographic Sheet

The demographic sheet was used to collect basic information about the participant and their sibling. The questions included: Age, Gender, Marital Status, Education level, Family type, Socio-economic Status, Area of residence, Parental marital status, Average Screen Time, Purpose of phone use, Total number of siblings, Birth Order, Age gap with sibling, Frequency of interaction with sibling, and Perceived closeness with sibling.

Generic Scale of Being Phubbed (GSBP)

GSBP was used to measure the experience of Being Phubbing by an individual, adapted to a sibling context. It was developed by Chotpitayasunondh & Douglas in 2018.

It is a brief scale, comprising of 22 items. There are three underlying factors on which the scale is based, *Perceived Norms (PN)*, reflects descriptions of what others do with their phones (1-9 items of the scale), *Feeling Ignored (FI)*, concerning feeling ignored by other's

phone use (10-17 items of the scale), and *Interpersonal Conflict (IC)*, concerning perceived conflict between oneself and others due to mobile phone use (18-22 items of the scale).

There are seven response categories in this scale which are: Never, Rarely, Occasionally, Sometimes, Frequently, Usually and Always. It has a score range from 1 to 7. The GSBP measure is scored by summing up all item scores. It has high internal consistency reliability with Cronbach's alpha of 0.96 (Ippolito, 2020).

Lifespan Sibling Relationship Scale (LSRS)

Lifespan Sibling Relationship Scale was used to measure the Sibling Relationship, through three dimensions of sibling relationship in childhood, retrospectively, and in adulthood: frequency and positivity of behavior towards the sibling, affect towards the sibling, and beliefs about the sibling and the sibling relationship. It was developed by Heidi R. Riggio in 2000.

It is a long scale, comprising of 48 items and 6 subscales, with 8 items for each subscale. The subscales are Adult Affect, Adult Behavior, Adult Cognitions, Child Affect, Child Behavior, and Child Cognitions. In this study, 3 subscales out of the total 6 subscales were used. The 3 subscales were: Adult Affect, Adult Behavior, and Adult Cognition.

There are five response categories, ranging from strongly agree to strongly disagree. The score ranges are 1 to 5. The First Subscale, Adult Affect, comprises of 1-8 items in the scale. Second Subscale, Adult Behaviour, comprises of 9-16 items in the scale and the Third Subscale, Adult Cognitions, consists of 17-24 items in the scale. Item 6, 13 and 19 are reverse scored items. The total score range for Adult Scales is 24-120, and for each subscale is 8-40.

It has high internal consistency reliability with Cronbach's alpha of 0.96 (Riggio, 2000). The six LSRS subscales also have high coefficient alphas (Adult Affect= 0.91, Adult Behavior=

0.87, Adult Cognitions= 0.91, Child Affect= 0.89, Child Behavior= 0.84, and Child Cognitions= .88) (Lin et al., 2024).

UCLA Loneliness Scale – Version 3

UCLA Loneliness Scale (Version 3) was used to measure loneliness perceived by individuals, adapted to sibling context. It was developed by Daniel Russell in 1996.

It consists of 20 items, with four response categories: Never, Rarely, Sometimes, and Always. To calculate the total score for each participant, all responses are summed up to obtain a score ranging from 20 to 80. Items 1, 5, 6, 9, 10, 15, 16, 19, and 20 are reverse-scored.

It has high internal consistency reliability with Cronbach's alpha ranging from 0.89 to 0.94 (Russell D. W., 1996).

Procedure

The process began with the building of a questionnaire for the purpose of data collection. The questionnaire consisted of informed consent, demographic questions, and three scales used to measure the study variables. All three scales including Generic Scale of Being Phubbed (GSBP), Lifespan Sibling Relationship Scale (LSRS), and UCLA Loneliness Scale were selected on the basis of item suitability for the research sample. Authors were approached via email to obtain permission for using the scale in our study where needed.

Prior to data collection process, written approval from Department of Clinical Psychology of Shifa Tameer-e-Millat University was taken. For data collection, purposive sampling technique was used and individuals were approached through friends, family, and university students. Paper based questionnaire was used to gather data. Before providing the questionnaire, individuals were asked a few questions to filter the participants and ensure fulfillment of the inclusion criteria. Those individuals who did not have a sibling above the age

of 18, were living in hostels, or had siblings with mental health or substance abuse problem were not included in the data collection process.

Informed consent was then obtained from eligible participants to ensure voluntary participation and anonymity. The time taken to fill each form was around 08–10 minutes. Data was gathered from a total of 260 participants which include 130 female and 130 male participants to ensure equal representation. Gathered data was entered into Statistical Package for Social Sciences (SPSS), using which data analysis was carried out.

Statistical Analyses

Before conducting the main statistical tests, several preliminary analyses were carried out to examine the characteristics and quality of the data. Reliability analysis was conducted to check the internal consistency of each scale used in the study. Skewness and kurtosis values were examined to assess whether the data met the assumption of normal distribution, which is required for the planned parametric tests.

Independent samples t-test and ANOVA were conducted to explore group differences based on gender and age of the participants simultaneously. Pearson product-moment correlation was conducted to examine the strength and direction of the relationship between the key variables. This allowed the study to determine whether higher levels of Being Phubbed were associated with lower Sibling Relationship Quality and greater Loneliness.

Simple linear regression analysis was used to explore whether being phubbed significantly predicted sibling relationship quality. This allowed examination of the directional relationship between the experience of being ignored due to smartphone use and the quality of the sibling bond. The second regression analysis tested whether Being Phubbed significantly predicted Loneliness, considering Loneliness as a second dependent variable. These analyses

helped to determine the extent to which being phubbed contributes to changes in sibling relationship quality and Loneliness.

Mediation analysis was conducted to examine whether Loneliness mediates the relationship between Being Phubbed and Sibling Relationship Quality and its subscales. It was performed using the PROCESS macro (Model 4) by Hayes (2022), version 4.2.

Ethical Considerations

The research was conducted in compliance to the APA ethical codes of conduct. Written approval from Department of Clinical Psychology of Shifa Tameer-e-Millat University was taken prior to conduction of research. Ethical principles were followed throughout the process of research. Informed consent was used to inform the participants about purpose of the study and their voluntary participation during data collection stage. Participants were assured that their responses will remain confidential and will only be used for academic purpose. Participants were also informed that they have the right to withdraw from the participation at any point without potential consequences. The study involved minimal risk, adhered to ethical research guidelines, and respected the dignity, privacy, and rights of all participants.

Chapter 3

Results

*Frequency and Percentage***Table 1***Frequency & Percentage of demographic variables (N=260)*

Variable	Category	<i>f</i>	%
<i>Age</i>	18-21	120	46.2
	22-25	118	5.4
	26-30	22	8.5
<i>Gender</i>	Male	130	50.0
	Female	130	50.0
<i>Education Level</i>	Undergraduate	203	78.1
	Graduate	47	18.1
	Post-graduate	10	3.8
<i>Socio Economic Status</i>	Lower	14	5.4
	Middle	220	84.6
	Upper	26	10.0
<i>Family type</i>	Nuclear	200	76.9
	Joint	60	23.1
<i>Residence</i>	Urban	206	79.2
	Rural	54	20.8
<i>Marital Status</i>	Single	231	88.8
	Married	21	8.1
	Other	8	3.1

Note. *f* = frequency, % = percentage

Above table displays the frequency and percentage distribution of demographic variables within the study. The table provides demographic information on age, gender, educational level, socioeconomic status, family type, residence and marital status. Among the participants, 46.2% belong to 18-21 age criteria, 5.4 to 22-25 age criteria and only 8.5% to the 25-30 age category. Both male and Women are equal in the study. 78.1% are undergraduates, 18.1% are graduates and 3.8% are post graduates. In terms of socioeconomic status, 5.4% are from lower class, 84.6% are from middle class and 10.0% are from upper class. Family type shows that 76.9% are from nuclear families whereas 23.2% are from joint families. 79.2% of the participants reside in urban areas where as 20.8% reside in rural areas. Moreover, 88.8% of participants are single, 8.1% are married and 3.1% belong to the “other” category which includes, engaged, divorced or a relationship status not captured by the single/married categories.

Psychometric Properties

Table 2

Psychometric Properties of Study Variable Scales (N=260)

Variables	k	A	M(SD)	Potential	Actual	Skewness	Kurtosis
<i>GSBP</i>	22	0.95	77.2 (30.2)	7-154	23-144	0.24	-0.99
<i>Total LSRS</i>	24	0.94	91 (18.9)	24-120	33-120	-0.66	-0.20
<i>LSRS-AA</i>	8	0.88	32 (6.44)	8-40	8-40	-1.05	0.96
<i>LSRS-AB</i>	8	0.86	28.8 (6.90)	8-40	11-40	-0.42	-0.67
<i>LSRS-AC</i>	8	0.89	30.1 (7.2)	8-40	9-40	-0.72	0.02
<i>LS</i>	20	0.87	48.1 (10.5)	20-80	22-77	-0.30	-0.16

Note. k= number of items, α = Cronbach's alpha, M = Mean, SD = Standard deviation, GSBP = Generic Scale of Being Phubbed, LSRS = Lifespan Sibling Relationship Scale; AA = Adult Affect, AB = Adult Behavior, AC = Adult Cognition, LS = Loneliness Scale

Table 2 provides descriptive statistics, alpha coefficients, and item counts for the Generic Scale of Being Phubbed, Lifespan Sibling Relationship Quality along with the subscales, and Loneliness Scale. The Cronbach's alpha coefficient value for Generic Scale of being Phubbed is 0.95 indicating excellent reliability. For Lifespan Sibling Relationship Quality alpha coefficient value is 0.94 indicating excellent reliability. The subscales Adult Affect has 0.88, Adult Behavior has 0.86, and Adult Cognition has 0.89 value indicating good reliability. Loneliness Scale has 0.87 alpha coefficient value indicating good reliability. The standard skewness and kurtosis values are falling within the range of -1 to +1, indicating that the data follows an approximate normal distribution, except Lifespan Sibling Relationship subscale, Adult Affect which shows a skewness value of -1.05 indicating a substantial negatively skewed distribution.

Correlation Analysis Between Being Phubbed, Sibling Relationship Quality and Loneliness

Table 3

Pearson product correlations for comparison of study variables (N=260)

Variables	GSBP	LSRS	LSRS-AA	LSRS-AB	LSRS-AC	LS
<i>GSBP</i>	-					
<i>Total LSRS</i>	-0.44**	-				
<i>LSRS-AA</i>	-0.34**	0.88**	-			
<i>LSRS-AB</i>	-0.40**	0.90**	0.65**	-		
<i>LSRS-AC</i>	-0.46**	0.95**	0.79**	0.82**	-	
<i>LS</i>	0.33**	-0.40**	-0.33**	-0.35**	-0.42**	-

Note: M = Mean, SD = Standard Deviation, CI = Confidence interval; LL = Lower limit, UL = Upper limit, GSBP = Generic Scale of Being Phubbed, LSRS = Lifespan Sibling Relationship Scale; AA = Adult Affect, AB = Adult Behavior, AC = Adult Cognition, LS = Loneliness Scale

The table presents Pearson product-moment correlation coefficients between six variables: Being Phubbed, Sibling Relationship Quality and its subscales; Adult Affect, Adult Behavior, Adult Cognition, and Loneliness. Being Phubbed (M=77.2, SD=30.2) shows a significant moderate negative correlation with Sibling Relationship Quality ($r = -0.44$, $p < 0.01$), Adult Affect ($r = -0.34$, $p < 0.01$), Adult Behavior ($r = -0.40$, $p < 0.01$), Adult Cognition ($r = -0.46$, $p < 0.01$), and a significant moderate positive relationship with Loneliness ($r = 0.33$, $p < 0.01$). This indicates that an increase in Being Phubbed is associated with a slight decrease in Sibling Relationship Quality and increase in Loneliness. Total Sibling Relationship Quality (M = 91, SD = 18.9) has very strong positive correlations with its subscales; Adult Affect ($r = 0.88$, $p < 0.01$), Adult Behavior ($r = 0.90$, $p < 0.01$), and Adult Cognition ($r = 0.95$, $p < 0.01$), and a significant moderate negative correlation with Loneliness ($r = -0.40$, $p < 0.01$). This indicates that an increase in Sibling Relationship scores is associated with higher Adult Affect, Adult Behavior and Adult Cognition levels in participants with a moderate decrease in Loneliness.

Among the subscales, Adult Affect ($M = 32$, $SD = 6.44$) shows a significant positive correlation with Adult Behavior ($r = 0.65$, $p < 0.01$), Adult Cognition ($r = 0.79$, $p < 0.01$) and significant moderate negative correlation with Loneliness ($r = -0.33$, $p < 0.01$). This indicates that an increase in Adult Affect level is associated with increase in Adult Behavior and Adult Cognition levels with a decrease in Loneliness. Adult Behavior subscale shows a very strong correlation with Adult Cognition ($r = 0.82$, $p < 0.01$) and significant moderate negative correlation with Loneliness ($r = -0.35$, $p < 0.01$). This indicates that increase in Adult Behavior scores is associated with increase in Adult Affect and decrease in Loneliness. Adult Cognition ($M = 30.1$, $SD = 7.2$) shows a significant moderate negative correlation with Loneliness ($r = -0.42$, $p < 0.01$). This indicates that increase in Adult Cognition is associated with decrease in Loneliness.

Linear Regression Analysis Predicting Sibling Relationship Quality from Being Phubbed

Table 4

Linear regression for prediction of study variables (N=260)

<i>Predictors</i>	<i>R</i>	<i>R</i> ²	<i>F</i>	<i>B</i>	<i>t</i>	<i>Confidence Interval</i> 95%	
						<i>LL</i>	<i>UL</i>
Criterion Variable= LSRS							
Model 1	0.44	0.196	63.09				
GSBP				-0.27	-7.94	-0.34	-0.20

Note. LSRS=Lifespan Sibling Relationship Scale, GSBP= Generic Scale of Being Phubbed, CI= Confidence Interval, LL= Lower Limit, UL= Upper Limit, F= Explained Variance, t= Mean Difference

The dependent variable, Sibling Relationship Quality, was regressed on the predicting variable, Being Phubbed, to test the research hypothesis. The model was significant ($F=63.09$, $B = -0.27$, $p < .001$,) indicating that Being Phubbed significantly predicts lower Sibling Relationship Quality. The R^2 value of 0.196 indicates that the predictor, Being Phubbed, explains 19.6% of the variance in Sibling Relationship Quality.

Linear Regression Analysis Predicting Loneliness from Being Phubbed

Table 5

Linear regression for prediction of study variables (N=260)

<i>Predictors</i>	<i>R</i>	<i>R</i> ²	<i>F</i>	<i>B</i>	<i>t</i>	<i>Confidence Interval</i> 95%	
						<i>LL</i>	<i>UL</i>
Criterion Variable= LS							
Model 1	0.33	0.112	32.67				
GSBP				0.11	5.71	0.07	0.15

Note. LS=Loneliness Scale, GSBP= Generic Scale of Being Phubbed, CI= Confidence Interval, LL= Lower Limit, UL= Upper Limit, R= Explained Variance, t= Mean Difference

The dependent variable, Loneliness, was regressed on the predicting variable, Being Phubbed, to test the research hypothesis. The model was significant ($F=32.67$, $B = 0.11$, $p < .001$), indicating that Being Phubbed significantly predicts higher Loneliness. The R^2 value of 0.112 indicates that the predictor, Being Phubbed, explains 11.2% of the variance in Loneliness.

Mediating Role of Loneliness between Being Phubbed and Adult Affect of Sibling Relationship Quality

Table 6

Mediating Effect of Loneliness in Predicting Adult Affect of Sibling Relationship Quality through Being Phubbed (N= 260)

<i>Model</i>	<i>B</i>	<i>SE</i>	<i>T</i>	<i>F</i>	<i>R</i>	<i>R</i> ²	<i>p</i>	<i>95 % CI</i>	
								<i>LL</i>	<i>UL</i>
Model 1: Being Phubbed → Loneliness									
				32.67	0.33	0.11	.000		
Constant	39.11	1.69	23.05				.000	35.77	42.45
BP→ LS	0.11	0.02	5.71				.000	0.07	0.15
Model 2: Being Phubbed & Loneliness → Adult Affect									
				36.50	0.41	0.17	.000		
Constant	43.58	1.75	24.80				.000	40.12	47.04
BP→ AA	-0.05	0.01	-4.27				.000	-0.08	-0.02
LS → AA	-0.15	0.03	-4.12				.000	-0.22	-0.07
Direct effect	-0.05	0.01	-4.27				.000	-0.08	-0.02
(c')									
Indirect effect	-0.01	0.00	-					-0.02	-.009
(a x b)									
Total effect (c)	-0.07	0.01	-5.82	33.97	0.34	0.11	.000	-0.09	-0.04

Note: BP= Being Phubbed, LS= Loneliness, AA= Adult Affect

The above table illustrates the mediating role of Loneliness in the relationship between Being Phubbed and Adult Affect of Sibling Relationship Quality among young adults.

Mediation analysis was performed using the PROCESS macro (Model 4) by Hayes (2022), version 4.2, to test the hypothesis.

Table 6 shows that the path (a) from Being Phubbed to Loneliness is significant and positive ($\beta = 0.11$, $t = 5.71$, $p = .000$, $CI = [.07, .15]$), accounting for 11.2% of the variance in Loneliness ($R^2 = 0.112$, $F = 32.67$, $p < 0.001$). This reflects a moderate effect of Being Phubbed on the mediator, Loneliness. Similarly, the path (b) from Loneliness to Adult Affect is significant and negative ($\beta = -0.15$, $t = -4.12$, $p = .000$, $CI = [-.22, -.07]$). Furthermore, the direct path (c') from Being Phubbed to Adult Affect remains significant and negative ($\beta = -0.05$, $t = -4.27$, $p = .000$, $CI = [-.08, -.02]$). The total effect (c) of Being Phubbed on Adult Affect, excluding the mediator Loneliness, is also negative and statistically significant ($\beta = -0.07$, $t = -5.82$, $p = .000$, $CI = [-.09, -.04]$).

The indirect effect of Being Phubbed on Adult Affect through Loneliness was negative and statistically significant ($\beta = -0.01$, $CI = [-.02, -.009]$), indicating that Loneliness serves as a partial mediator in the relationship between Being Phubbed and Adult Affect of Sibling Relationship Quality. The mediation model accounted for 11% of the variance in Adult Affect ($R^2 = 0.11$), and the persistence of a significant direct effect suggests partial rather than full mediation, reflecting a modest yet meaningful explanatory contribution of Loneliness.

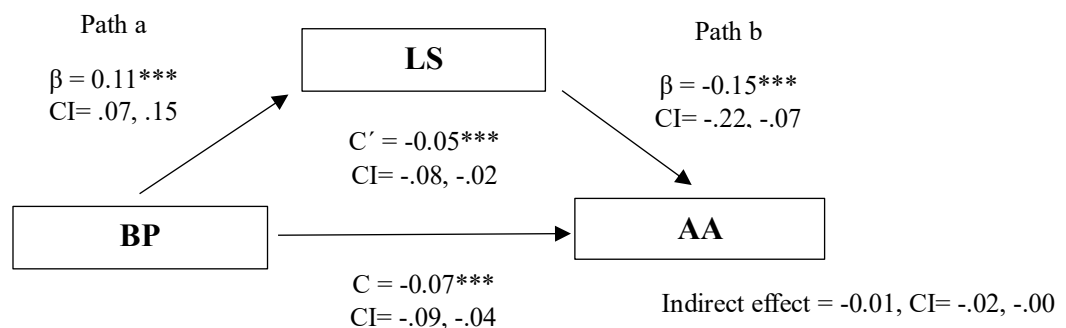


Figure 2. Impact of Loneliness on Adult Affect of Sibling Relationship Quality through Being Phubbed.

Mediating Role of Loneliness between Being Phubbed and Adult Behaviour of Sibling Relationship Quality

Table 7

Mediating Effect of Loneliness in Predicting Adult Behaviour of Sibling Relationship Quality through Being Phubbed (N= 260)

<i>Model</i>	<i>B</i>	<i>SE</i>	<i>T</i>	<i>F</i>	<i>R</i>	<i>R</i> ²	<i>P</i>	<i>95 % CI</i>	
								<i>LL</i>	<i>UL</i>
Model 1: Being Phubbed → Loneliness									
				32.67	0.33	0.11	.000		
Constant	39.11	1.69	23.05				.000	35.77	42.45
BP→ LS	0.11	0.02	5.71				.000	0.07	0.15
Model 2: Being Phubbed & Loneliness → Adult Behaviour									
				36.50	0.47	0.22	.000		
Constant	42.45	1.82	23.27				.000	38.85	46.04
BP→ AB	-0.07	0.01	-5.51				.000	-0.09	-0.04
LS → AB	-0.16	0.03	-4.29				.000	-0.23	-0.08
Direct effect	-0.07	0.01	-5.51				.000	-0.09	-0.04
(c´)									
Indirect effect	-0.01	0.00	-					-0.03	-.009
(a x b)									
Total effect (c)	-0.09	0.01	-7.14	51.07	0.40	0.16	.000	-0.11	-0.06

Note: BP= Being Phubbed, LS= Loneliness, AB= Adult Behaviour

The above table illustrates the mediating role of Loneliness in the relationship between Being Phubbed and Adult Behaviour of Sibling Relationship Quality among young adults.

Mediation analysis was performed using the PROCESS macro (Model 4) by Hayes (2022), version 4.2, to test the hypothesis.

Table 7 shows that the path (a) from Being Phubbed to Loneliness is significant and positive ($\beta = 0.11$, $t = 5.71$, $p = .000$, $CI = [.07, .15]$), accounting for 11.2% of the variance in Loneliness ($R^2 = 0.112$, $F = 32.67$, $p < 0.001$). This reflects a moderate effect of Being Phubbed on the mediator, Loneliness. Similarly, the path (b) from Loneliness to Adult Behaviour is significant and negative ($\beta = -0.16$, $t = -4.29$, $p = .000$, $CI = [-.23, -.08]$). Furthermore, the direct path (c') from Being Phubbed to Adult Behaviour remains significant and negative ($\beta = -0.07$, $t = -5.51$, $p = .000$, $CI = [-.09, -.04]$). The total effect (c) of Being Phubbed on Adult Behaviour, excluding the mediator Loneliness, is also negative and statistically significant ($\beta = -0.09$, $t = -7.14$, $p = .000$, $CI = [-.11, -.06]$).

The indirect effect of Being Phubbed on Adult Behaviour through Loneliness was negative and statistically significant ($\beta = -0.01$, $CI = [-.03, -.009]$), indicating that Loneliness partially mediates the relationship between Being Phubbed and Adult Behaviour of Sibling Relationship Quality. The mediation model explained 16% of the variance in Adult Behaviour ($R^2 = 0.16$), and the continued significance of the direct effect supports partial mediation, suggesting a modest yet meaningful predictive contribution of Loneliness.

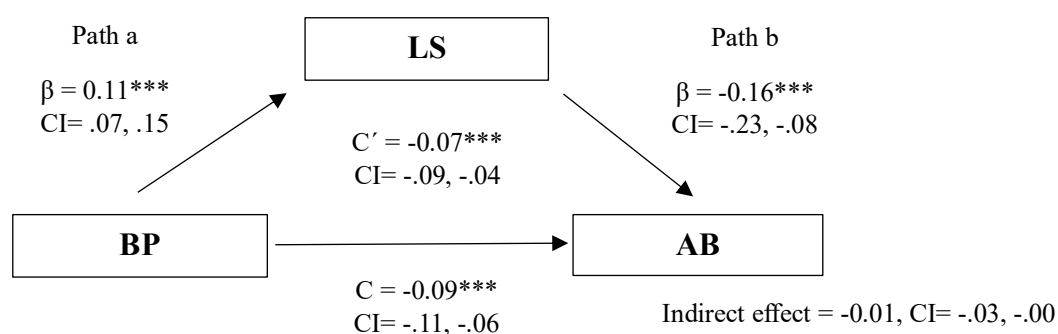


Figure 3. Impact of Loneliness on Adult Behaviour of Sibling Relationship Quality through Being Phubbed.

Mediating Role of Loneliness between Being Phubbed and Adult Cognition of Sibling Relationship Quality

Table 8

Mediating Effect of Loneliness in Predicting Adult Cognition of Sibling Relationship Quality through Being Phubbed (N= 260)

<i>Model</i>	<i>B</i>	<i>SE</i>	<i>T</i>	<i>F</i>	<i>R</i>	<i>R</i> ²	<i>P</i>	<i>95 % CI</i>	
								<i>LL</i>	<i>UL</i>
Model 1: Being Phubbed → Loneliness									
				32.67	0.33	0.11	.000		
Constant	39.11	1.69	23.05				.000	35.77	42.45
BP→ LS	0.11	0.02	5.71				.000	0.07	0.15
Model 2: Being Phubbed & Loneliness→ Adult Cognition									
				55.12	0.54	0.30	.000		
Constant	46.99	1.81	25.89				.000	43.42	50.57
BP→ AC	-0.08	0.01	-6.56				.000	-0.11	-0.06
LS→ AC	-0.21	0.03	-5.51				.000	-0.28	-0.13
Direct effect	-0.08	0.01	-6.56				.000	-0.11	-0.06
(c')									
Indirect effect	-0.02	0.00	-					-0.03	-0.01
(a x b)									
Total effect (c)	-0.11	0.01	-8.46	71.65	0.46	0.21	.000	-0.13	-0.08

Note: BP= Being Phubbed, LS= Loneliness, AC= Adult Cognition

The above table illustrates the mediating role of Loneliness in the relationship between Being Phubbed and Adult Cognition of Sibling Relationship Quality among young adults.

Mediation analysis was performed using the PROCESS macro (Model 4) by Hayes (2022), version 4.2, to test the hypothesis.

Table 8 shows that the path (a) from Being Phubbed to Loneliness is significant and positive ($\beta = 0.11$, $t = 5.71$, $p = .000$, $CI = [.07, .15]$), accounting for 11.2% of the variance in Loneliness ($R^2 = 0.112$, $F = 32.67$, $p < 0.001$). This reflects a moderate effect of Being Phubbed on the mediator, Loneliness. Similarly, the path (b) from Loneliness to Adult Cognition is significant and negative ($\beta = -0.21$, $t = -5.51$, $p = .000$, $CI = [-.28, -.13]$). Furthermore, the direct path (c') from Being Phubbed to Adult Cognition remains significant and negative ($\beta = -0.08$, $t = -6.56$, $p = .000$, $CI = [-.11, -.06]$). The total effect (c) of Being Phubbed on Adult Cognition, excluding the mediator Loneliness, is also negative and statistically significant ($\beta = -0.11$, $t = -8.46$, $p = .000$, $CI = [-.13, -.08]$).

The indirect effect of Being Phubbed on Adult Cognition through Loneliness was negative and statistically significant ($\beta = -0.02$, $CI = [-.03, -.01]$), indicating that Loneliness partially mediates the relationship between Being Phubbed and Adult Cognition of Sibling Relationship Quality. The mediation model accounted for 21% of the variance in Adult Cognition ($R^2 = 0.21$), and the persistence of a significant direct effect supports partial mediation, reflecting a modest yet meaningful explanatory contribution of Loneliness.

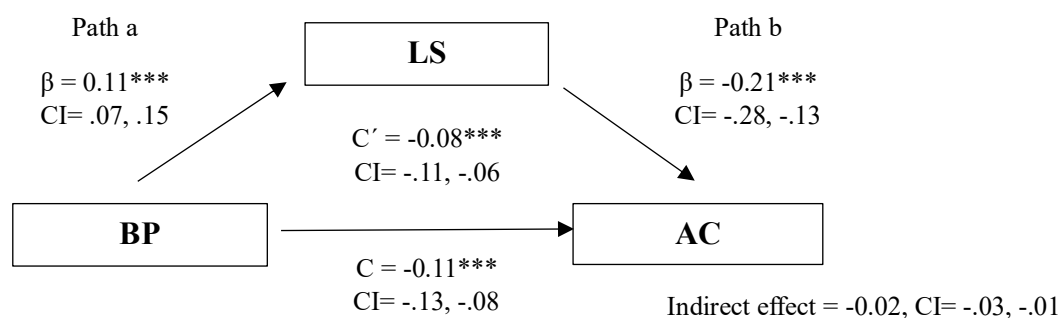


Figure 4. Impact of Loneliness on Adult Cognition of Sibling Relationship Quality through Being Phubbed.

Mediating Role of Loneliness between Being Phubbed and Sibling Relationship Quality

Table 9

Mediating Effect of Loneliness in Predicting Sibling Relationship Quality through Being Phubbed (N= 260)

<i>Model</i>	β	<i>SE</i>	<i>T</i>	<i>F</i>	<i>R</i>	<i>R</i> ²	<i>P</i>	<i>95 % CI</i>	
								<i>LL</i>	<i>UL</i>
Model 1: Being Phubbed → Loneliness									
				32.67	0.33	0.112	.000		
Constant	39.11	1.69	23.05				.000	35.77	42.45
BP→ LS	0.11	0.02	5.71				.000	0.07	0.15
Model 2: Being Phubbed & Loneliness → Sibling Relationship Quality									
				48.24	0.52	0.273	.000		
Constant	133.03	4.82	27.55				.000	123.5	142.5
BP→ SRQ	-0.21	0.03	-6.10				.000	-0.28	-0.14
LS→ SRQ	-0.52	0.10	-5.19				.000	-0.72	-0.32
Direct effect	-0.21	0.03	-6.10				.000	-0.28	-0.14
(c´)									
Indirect effect	-0.06	0.01	-					-0.09	-0.03
(a x b)									
Total effect (c)	-0.27	0.03	-7.94	63.09	0.44	0.196	.000	-0.34	-0.20

Note: BP= Being Phubbed, LS= Loneliness, SRQ= Sibling Relationship Quality

The above table illustrates the mediating role of Loneliness in the relationship between Being Phubbed and Sibling Relationship Quality among young adults. Mediation analysis was

performed using the PROCESS macro (Model 4) by Hayes (2022), version 4.2, to test the hypothesis.

Table 9 shows that the path (a) from Being Phubbed to Loneliness is significant and positive ($\beta = 0.11$, $t = 5.71$, $p = .000$, $CI = [.07, .15]$), accounting for 11.2% of the variance in Loneliness ($R^2 = 0.112$, $F = 32.67$, $p < 0.001$). This reflects a moderate effect of Being Phubbed on the mediator, Loneliness. Similarly, the path (b) from Loneliness to Sibling Relationship Quality is significant and negative ($\beta = -0.52$, $t = -5.19$, $p = .000$, $CI = [-.72, -.32]$). Furthermore, the direct path (c') from Being Phubbed to Sibling Relationship Quality remains significant and negative ($\beta = -0.21$, $t = -6.10$, $p = .000$, $CI = [-.28, -.14]$). The total effect (c) of Being Phubbed on Sibling Relationship Quality, excluding the mediator Loneliness, is also negative and statistically significant ($\beta = -0.27$, $t = -7.94$, $p = .000$, $CI = [-.34, -.20]$).

The indirect effect of Being Phubbed on Sibling Relationship Quality through Loneliness was negative and significant ($\beta = -0.06$, $CI = [-.09, -.03]$), indicating that Loneliness partially mediates the relationship between Being Phubbed and Sibling Relationship Quality. Since the total variance accounted for by the model was 19.6% ($R^2 = 0.196$), and the direct effect did not reduce, this provides evidence for partial mediation by Loneliness in the relationship between Being Phubbed and Sibling Relationship Quality, suggesting a modest yet meaningful predictive value of the mediation model.

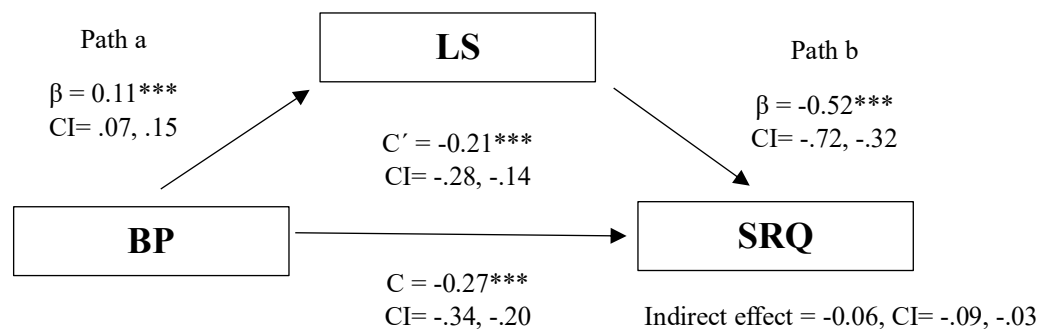


Figure 5. Impact of Loneliness on Sibling Relationship Quality through Being Phubbed.

*Independent Sample T-test between Gender***Table 10***Group difference between gender of participants (N=260)*

	Men	Women			95% CI		
	n = (130)	n = (130)					
Variables	M(SD)	M(SD)	t(df)	P	LL	UL	Cohen's d
<i>GSBP</i>	85.8(30.2)	68.6(27.7)	4.7(258)	.000	10.0	24.1	0.59
<i>Total LSRS</i>	84.7(17.8)	97.3(17.8)	-5.6(258)	.000	-16.9	-8.2	-0.5
<i>LSRS-AA</i>	30.4(6)	33.6(6.4)	-4(258)	.000	-4.6	-1.6	-0.52
<i>LSRS-AB</i>	26.3(6.6)	31.3(6.1)	-6.3(258)	.000	-6.6	-3.4	-0.79
<i>LSRS-AC</i>	27.9(7)	32.3(6.7)	-5.1(258)	.000	-6.0	-2.6	-0.64
<i>LS</i>	50.1(10)	46.1(10.6)	-3.1(388)	.002	1.5	6.6	0.39

Note: M = Mean, SD = Standard Deviation, CI = Confidence interval; LL = Lower limit, UL = Upper limit, GSBP = Generic Scale of Being Phubbed, LSRS = Lifespan Sibling Relationship Scale; AA = Adult Affect, AB = Adult Behavior, AC = Adult Cognition, LS = Loneliness Scale

The above table compares Being phubbed, Sibling relationship quality, and Loneliness among men and Women. Men scored significantly higher than Women on being phubbed while Women scored higher than Men on all three subscales of sibling relationship quality, namely Affect, Behavior, and Cognition. Loneliness also showed significant gender differences. The results indicate clear gender differences in sibling dynamics. Men scored significantly higher on being phubbed ($M = 85.8$, $SD = 30.2$) compared to Women ($M = 68.6$, $SD = 27.7$), $t(258) = 4.7$, $p < .001$, $d = 0.59$, suggesting that Men experience more phubbing

from siblings. Women scored higher on all subscales of sibling relationship quality—Affect ($M = 33.6$ vs. 30.4 , $t(258)$), Behavior ($M = 31.3$ vs. 26.3 , $t(258)$) and Cognition ($M = 32.3$ vs. 27.9 , $t(258)$). Men also reported higher loneliness ($M = 50.1$ vs. 46.1 , $t(388)$).

Chapter 4

Discussion

The current study aimed to examine the relationship between Being Phubbed by a sibling, Sibling Relationship Quality, and Loneliness among young adults. Several data analyses were conducted to test the research hypotheses, including correlation, t-test for group gender differences, ANOVA for group age differences, regression models, and mediation analysis. The findings provide evidence that the experience of being phubbed in a sibling relationship dynamic plays a significant role in shaping the relationship quality and feelings of loneliness in an individual.

The study found that Being Phubbed by a sibling was significantly, moderately and negatively correlated with overall Sibling Relationship Quality and its sub-dimensions—Affect, Behaviour, and Cognition. Regression analysis indicated that Being Phubbed by a sibling significantly predicted lower sibling relationship quality, explaining 19.6% of its variance. These results support first and second hypothesis of the study, indicating that when siblings are frequently ignored due to smartphone use, they perceive the relationship to be less warm, less supportive, and less communicative. These findings are consistent with research done by Robert and David (2016) which found that being ignored due to smartphone use decreases perceived relationship satisfaction.

Further statistical analyses in the study found that a significant moderate positive correlation emerged between Being Phubbed and Loneliness. When siblings engage in phubbing behavior, it reduces the emotional intimacy and quality of communication among the siblings which potentially contributes to feelings of loneliness (Ergün et al., 2020; Stocker et al., 2020). Regression results also showed that Being Phubbed significantly predicted higher

loneliness, explaining 11.2% of its variance. This finding supports third hypothesis of the study, indicating that the experience of being phubbed contributes to feelings of loneliness.

Mediation analysis indicated that Loneliness partially mediates the relationship between Being Phubbed and Sibling Relationship Quality. The indirect effect was significant, implying that being phubbed by a sibling reduces sibling relationship quality partly because it increases feelings of loneliness. In addition, when considered separately in sub-dimensions of sibling relationship quality, loneliness was identified as a partial mediator in the association between being phubbed and adult affect, behaviour, and cognition. Being phubbed was linked to greater loneliness, which predicted less emotional intimacy, more negative sibling relations and more negative cognitive appraisals of the sibling relationship. The mediation was only partial because the direct effect was significant as well, indicating that phubbing harms sibling relationships both directly and indirectly through this pathway of loneliness. Research found that warmth between siblings was negatively associated with loneliness. Loneliness partially mediated the relationship between sibling relationship quality and well-being (Stocker et al., 2020). This supports fourth hypothesis of the study and suggests that when people feel neglected by their siblings, they internalize it and feel lonely, which further negatively affects their relationship. Having a positive and supportive relationship with the sibling can serve as a buffer against loneliness and emotional distress (Xu et al., 2022; Perlman & Peplau, 1982).

Independent sample t-tests revealed significant gender differences: Male participants scored higher on Being Phubbed and Loneliness while female participants scored higher on all dimensions of Sibling Relationship Quality. The finding that Women maintain higher sibling relationship quality may reflect cultural socialization of warmth, caretaking, and emotional expressiveness among women in South Asian societies. In contrast, high loneliness and phubbing among Men could reflect reduced communication patterns, higher device

engagement, or fewer emotional outlets (Ilyas et al., 2019). These results support the fifth hypothesis of the study, demonstrating that gender plays an important role in how phubbing is experienced and how sibling relationships are maintained.

ANOVA conducted to explore differences across various age groups revealed that Being Phubbed and Sibling Relationship Quality do not vary significantly by age while Loneliness varies significantly and is found to be highest in the age bracket 18-21. Consistent with our results, research has found that emerging adults report higher levels of loneliness than older adults indicating developmental differences in social connectedness and perceived isolation (Furmańska et al. 2025).

Overall, the results are in line with the previous literature that identifies phubbing as a relational disruptor. Previous studies have underlined that phubbing decreases perceived relational satisfaction, emotional support, and communication quality. This finding aligns with research which shows that phubbing leads to feelings of exclusion which then reduces relationship satisfaction (Al-Saggaf, 2022; Przybylski & Weinstein, 2013). Research on romantic partners and friends has also found out that being phubbed fosters emotional distance.

The current results also align with researches which indicate that phubbing predicts loneliness, poor relationship cohesion, and conflict. The mediation pattern is also consistent with previous models suggesting that emotional disconnection (loneliness) might be an important mechanism through which phubbing affects interpersonal relationships. When individuals feel replaced by technology during in-person interactions, it can reduce the sense of belonging and contribute to emotional distress (Baumeister & Leary, 1995; Hales et al., 2018). Furthermore, lack of meaningful connections can also contribute to feelings of loneliness and social isolation (Amichai-Hamburger & Etgar, 2016).

While several of these studies have investigated this topic, the present research extends literature by focusing on sibling relationships, which have received considerably less empirical attention despite the importance of sibling bonds in collectivist cultures like Pakistan. In Pakistan, siblings serve as primary sources of emotional support, companionship, and socialization within its collectivistic family system. For this reason, the act of phubbing in this relationship can be especially harmful (Fatima & Noor, 2022).

Suggestions

Future research should consider including both members of the sibling dyad as participants in order to capture a more comprehensive understanding of phubbing behaviours within sibling relationships. Assessing both self-reported phubbing and experiences of being phubbed from each sibling would allow for comparison of perspectives and reduce reliance on single-informant data. In addition, incorporating objective measures of phone use, such as screen-time tracking applications or digital monitoring tools, would provide more accurate indicators of phubbing behaviour and help minimize biases associated with self-report methods, including social desirability and recall errors. Using a dyadic and multi-method approach would strengthen the validity of findings and offer deeper insight into how phubbing operates within sibling interactions.

Upcoming researches may explore additional mediating or moderating variables that influence the relationship between phubbing and sibling relationship quality. Variables such as attachment style, emotional intelligence, empathy, conflict resolution skills, or self-esteem could provide deeper understanding of why some individuals are more affected by phubbing than others.

Researches in the future may consider using a longitudinal method to evaluate the change of phubbing, loneliness and sibling relationship quality in a period of time. A

longitudinal design would provide stronger evidence for causal relationships between being phubbed and loneliness or relationship quality than the cross-sectional design that was used in the current study because it could show a better understanding of whether phubbing increases loneliness or reduces sibling relationship quality over time.

Future research may explore differences in experiences amongst co-resident vs non-co-resident siblings, as the current study focused on sibling pairs living together. Studying sibling pairs living apart due to work, marriage, or education may highlight whether the impact of phubbing varies when face-to-face communication is limited and digital communication becomes more central.

It is also suggested that future research may want to consider using qualitative research techniques (e.g. interviews or focus groups) in order to obtain a more detailed and personal view of sibling phubbing experiences. In this case study, the use of qualitative analysis could yield information about how siblings interpret these experiences emotionally, how they are communicated and how these interact with other relational themes that would be difficult to assess with more traditional questionnaire methods.

Limitations

1. One of the limitations of the present study is its dependence on self-report questionnaires, which rely on participants' honesty and memory. Individuals may overestimate or underestimate the behaviour of phubbing and closeness to their sibling. Responses may also be influenced by social desirability, meaning participants might have answered in ways they believe answers are expected rather than with complete accuracy.
2. Another limitation is that the study did not take into account other important family or individual factors such as attachment style, personality traits or general communication

patterns that could influence how phubbing impacts sibling relationships. Without accounting for such variables, it is difficult to know whether the findings uncovered in the study are solely due to phubbing or partly influenced by these unmeasured factors.

3. Moreover, the sample was dominated by young adults between the ages of 18 and 21, mostly from urban areas and also from middle-class backgrounds. This concentration might have biased the participants' experiences and views about digital behaviour and sibling interactions, hence limiting the generalization to older groups, people in rural areas, or those from other socioeconomic settings.

Implications

1. One implication of the present study relates to family and sibling interactions. The findings suggest that phubbing can negatively impact sibling relationship quality by creating feelings of emotional distance and neglect. This highlights the importance of mindful phone use within the family, particularly during shared time with siblings, as excessive phone use may reduce meaningful communication and emotional closeness.
2. Another implication of this study is related to understanding relational dynamics among adult siblings. The mediating role of loneliness indicates that when siblings experience phubbing, they may feel ignored or disconnected, which can contribute to loneliness and, in turn, weaken the sibling bond. Families may benefit from being aware of how everyday digital behaviours can influence emotional experiences within sibling relationships.
3. The findings also imply that phone use during sibling interaction, such as time spent together within the family, may influence feelings of loneliness and relationship quality. Being more attentive during these interactions may support healthier and more connected sibling relationships.

4. The findings also have theoretical implications by extending existing phubbing research to the context of sibling relationships, which has received limited attention in previous literature. By focusing on siblings rather than romantic or peer relationships, the present study adds to the understanding of how digital behaviours operate within family systems and affect relationship quality.
5. The observed gender differences in phubbing and sibling relationship quality point toward the need for greater awareness of how digital behaviours may vary between Men and Women within families. Understanding these differences can help families reflect on communication patterns and encourage more balanced and respectful interactions between siblings.
6. Lastly, the findings of the present study offer a basis for future research on digital behaviour within sibling relationships. Future studies may further examine how phubbing and feelings of loneliness interact across different family settings or cultural contexts, which may help develop a clearer understanding of sibling dynamics in the digital age.

Conclusion

This study highlights the significant impact of being phubbed, by a sibling, on sibling relationship quality and loneliness among young adults. The findings suggest that being phubbed by a sibling result in lower quality of sibling relationship and contribute to feelings of loneliness, emphasizing the importance of appropriate, mindful digital behaviours. This research has placed the focus on sibling relationships, which have previously received little attention and, within a collectivist culture like Pakistan, are considered to hold the central position in the emotional support and social connectedness of the individual. Overall, these

findings point out the need to raise awareness regarding technology use and its subtle yet meaningful effects on our closest relationships.

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Appendices

Appendix A

Permission Letter for Data Collection



**Department of Clinical
Psychology**



September 26, 2025

To Whom It May Concern

Subject: REQUEST FOR DATA COLLECTION

Respected Sir/Madam,
Assalam-O-Alaikum

This is to certify that **Ms. Maheen Sohail**, a student of BS Clinical Psychology, 8th Semester (Session Spring 2022), Registration No. BS(CP)-22S-0204, at the Department of Clinical Psychology, Shifa Tameer-e-Millat University (STMU), is seeking permission for academic attachment at your esteemed organization.

Her approved research topic is: **"When Siblings Drift Apart: Influence Of Being Phubbed On Siblings Relationship Quality And Loneliness Among Young Adults."**

STMU, being a federally chartered university, is committed to imparting quality education and preparing skilled professionals in diverse fields. As part of the academic curriculum, research engagement is essential for the professional development and personal growth of our students.

We kindly request your permission to allow **Ms. Maheen Sohail** to collect data for her research during the period **October 2025 to December 2025**. We believe this collaboration will help strengthen professional ties and contribute to meaningful academic exchange. Please be assured that all information collected will remain strictly confidential and will be used solely for research purposes.

For further details, you may contact **Ms. Fiza Rehman**, In-charge BSCP Program, Department of Clinical Psychology, STMU, at fiza.dcp@stmu.edu.pk.

We look forward to your kind cooperation.


Department of Clinical Psychology
Shifa Tameer-e-Millat University (STMU)
Sector H-8/4 Islamabad Pakistan

Prof. Dr. Tamkeen Saleem
Head of the Department
Department of Clinical Psychology
Shifa Tameer-e-Millat University
Islamabad.

Shifa Tameer-e-Millat University, Department of Clinical Psychology,
Park Road Campus, Islamabad. 051-8840615



Department of Clinical Psychology



September 26, 2025

To Whom It May Concern

Subject: REQUEST FOR DATA COLLECTION

Respected Sir/Madam,
Assalam-O-Alaikum

This is to certify that **Ms. Mariya Aman**, a student of BS Clinical Psychology, 8th Semester (Session Spring 2022), Registration No. **BS(CP)-22S-0205**, at the Department of Clinical Psychology, Shifa Tameer-e-Millat University (STMU), is seeking permission for academic attachment at your esteemed organization.

Her approved research topic is: **"When Siblings Drift Apart: Influence Of Being Phubbed On Siblings Relationship Quality And Loneliness Among Young Adults."**

STMU, being a federally chartered university, is committed to imparting quality education and preparing skilled professionals in diverse fields. As part of the academic curriculum, research engagement is essential for the professional development and personal growth of our students.

We kindly request your permission to allow Ms. Mariya Aman to collect data for her research during the period **October 2025 to December 2025**. We believe this collaboration will help strengthen professional ties and contribute to meaningful academic exchange. Please be assured that all information collected will remain strictly confidential and will be used solely for research purposes.

For further details, you may contact **Ms. Fiza Rehman**, In-charge BSCP Program, Department of Clinical Psychology, STMU, at fiza.dcp@stmu.edu.pk.

We look forward to your kind cooperation.

Department of Clinical Psychology
Shifa Tameer-e-Millat University (STMU)
Sector H-8/4 Islamabad Pakistan

Prof. Dr. Tamkeen Saleem
Head of the Department
Department of Clinical Psychology
Shifa Tameer-e-Millat University
Islamabad.

Shifa Tameer-e-Millat University, Department of Clinical Psychology,
Park Road Campus, Islamabad. 051-8840615



Department of Clinical Psychology



September 26, 2025

To Whom It May Concern

Subject: REQUEST FOR DATA COLLECTION

Respected Sir/Madam,
Assalam-O-Alaikum

This is to certify that **Ms. Mutaal Tariq**, a student of BS Clinical Psychology, 8th Semester (Session Spring 2022), Registration No. **BS(CP)-22S-0206**, at the Department of Clinical Psychology, Shifa Tameer-e-Millat University (STMU), is seeking permission for academic attachment at your esteemed organization.

Her approved research topic is: **"When Siblings Drift Apart: Influence Of Being Phubbed On Siblings Relationship Quality And Loneliness Among Young Adults."**

STMU, being a federally chartered university, is committed to imparting quality education and preparing skilled professionals in diverse fields. As part of the academic curriculum, research engagement is essential for the professional development and personal growth of our students.

We kindly request your permission to allow Ms. Mutaal Tariq to collect data for her research during the period **October 2025 to December 2025**. We believe this collaboration will help strengthen professional ties and contribute to meaningful academic exchange. Please be assured that all information collected will remain strictly confidential and will be used solely for research purposes.

For further details, you may contact **Ms. Fiza Rehman**, In-charge BSCP Program, Department of Clinical Psychology, STMU, at fiza.dcp@stmu.edu.pk.

We look forward to your kind cooperation.

Department of Clinical Psychology
Shifa Tameer-e-Millat University - STMU
Sector H-8/4 - Islamabad

Prof. Dr. Tamkeen Saleem
Head of the Department
Department of Clinical Psychology
Shifa Tameer-e-Millat University
Islamabad.

Appendix B

Ethical Approval Letter



Department of Clinical Psychology



Ethical Approval

DCP-EAL No:0069-BS
Maheen Sohail
BS(CP)- 22S-0264
Department of Clinical Psychology
Shifa Tameer e Millat University
Islamabad, Pakistan

September 22, 2025

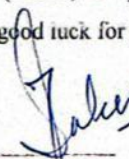
Dear Student,

On the behalf of the Preliminary Ethics Research Committee (PERC), I am pleased to confirm a favorable ethical opinion for your research under the supervision of Dr. Syeda Ayat E Zainab, titled **"When Siblings Drift Apart: Influence Of Being Phubbed On Siblings Relationship Quality And Loneliness Among Young Adults"** on the basis described in application form & supporting documents (application form, proposal, informed consent, study tool). I am pleased to inform that you may conduct your study. However, this approval is limited to the activities which are mentioned in the study. As part of this study, committee authorize the researcher to conduct study for degree purpose. You will conduct the study according to ethical protocols which are approved by American Psychiatric Association.

In case, if changes are required in any domain of your study (changes in title, data collection procedure, measures) you need to apply through proper channel to Preliminary Ethics Research Committee (PERC) for approval purpose, prior to implementation of changes.

I wish you good luck for your research.

Sincerely,


Chairperson (PERC)
Prof. Dr. Tankeen Saleem
Department of Clinical Psychology
Shifa Tameer e Millat University
Email: hod.dcp@dtmu.edu.pk



Department of Clinical Psychology



Ethical Approval

DCP-EAL No:00087B2
Mutaal Tariq
BS(CP)- 22S-0206
Department of Clinical Psychology
Shifa Tameer e Millat University
Islamabad, Pakistan

September 22, 2025

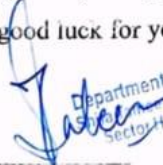
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I wish you good luck for your research.

Sincerely,


Department of Clinical Psychology
Shifa Tameer e Millat University (STMU)
Sector H-94 Islamabad Pakistan

Chairperson (PERC)
Prof.Dr. Tamkeen Saleem
Department of Clinical Psychology
Shifa Tameer e Millat University
Email: hod.dcp@dtmu.edu.pk



Department of Clinical Psychology



Ethical Approval

DCP-EAL. No.00010/BS

September 22, 2025

Mariya Aman

BS(CP)- 22S-0205

Department of Clinical Psychology

Shifa Tameer e Millat University

Islamabad, Pakistan

Dear Student,

On the behalf of the Preliminary Ethics Research Committee (PERC), I am pleased to confirm a favorable ethical opinion for your research under the supervision of Dr.Syeda Ayat E Zainab, titled **"When Siblings Drift Apart: Influence Of Being Phubbed On Siblings Relationship Quality And Loneliness Among Young Adults"** on the basis described in application form & supporting documents (application form, proposal, informed consent, study tool). I am pleased to inform that you may conduct your study. However, this approval is limited to the activities which are mentioned in the study. As part of this study, committee authorize the researcher to conduct study for degree purpose. You will conduct the study according to ethical protocols which are approved by American Psychiatric Association.

In case, if changes are required in any domain of your study (changes in title, data collection procedure, measures) you need to apply through proper channel to Preliminary Ethics Research Committee (PERC) for approval purpose, prior to implementation of changes.

I wish you good luck for your research.

Sincerely,

 Department of Clinical Psychology
Shifa Tameer-e-Millat University (STMU)
Sector H-8-4, Islamabad Pakistan

Chairperson (PERC)

Prof.Dr. Tankeen Saleem

Department of Clinical Psychology

Shifa Tameer e Millat University

Email: hod.dep@dtmu.edu.pk

Appendix C

Author's Consent to Use Scales



Karen Douglas <K.Douglas@kent.ac.uk>
to me ▾

Tue, Aug 12, 4:34 PM ☆ 😊 ↩ ⋮

Hi Mutaal,

Thank you for getting in touch. Please find the information attached and I wish you all the best with your research.

Karen.

Karen Douglas | Professor of Social Psychology (she/her)

School of Psychology, University of Kent, Canterbury, Kent CT2 7NP, United Kingdom

E-mail: k.douglas@kent.ac.uk

Internet: <https://www.kent.ac.uk/psychology/people/221/douglas-karen>

Principal Investigator "CONSPIRACY_FX: Consequences of conspiracy theories", ERC Advanced Grant

E-mail: conspiracyfx@kent.ac.uk

Internet: <https://research.kent.ac.uk/conspiracy-fx/>

From: Mutaal Tariq <mutaaltariq2003@gmail.com>

Date: Tuesday, 12 August 2025 at 11:37

To: vc216@kent.ac.uk <vc216@kent.ac.uk>, Karen Douglas <K.Douglas@kent.ac.uk>

Subject: Request for Permission to Use **GSPB** in Bachelor's Thesis

Appendix D

Consent Form

Study Title: When siblings drift apart: Influence of being phubbed on sibling relationship quality and loneliness among young adults.

This research is being conducted by BS Clinical Psychology students at Shifa Tameer-e-Millat University, Islamabad. The study is titled “When siblings drift apart: Influence of being phubbed on sibling relationship quality and loneliness among young adults.” It aims to explore how phubbing (using a mobile phone while ignoring a sibling) impacts sibling relationship quality and whether feelings of loneliness play a role. The focus is on siblings aged 18 above who live together.

Your participation is completely voluntary, you will be asked to fill out a questionnaire consisting of demographic information and standardized scales on phubbing, loneliness, and sibling relationship quality. Completing it will take around 5–10 minutes. You are free to withdraw at any point without giving a reason, and there will be no penalty for withdrawing.

There are no known risks associated with this study beyond those of daily life. While there may be no direct personal benefits, your responses will contribute to understanding how mobile phone use influences sibling relationships and emotional well-being. All information you provide will remain anonymous and confidential, used only for academic research purposes. No identifying details will be collected.

By proceeding with the survey, you confirm that you are 18 years of age or older, have read and understood the above information, and voluntarily agree to participate.

- ☐ I agree to participate.
- ☐ I do not agree to participate.

Appendix E

Demographic Sheet

Participant's Information

Age: _____

Gender: ☐ Male ☐ Female

Marital Status: ☐ Single ☐ Married

Education Level: ☐ Undergraduate ☐ Graduate ☐ Postgraduate

Family Type: ☐ Nuclear ☐ Joint

Residence: ☐ Urban ☐ Rural

Socioeconomic Status: ☐ Lower ☐ Middle ☐ Upper

Average Screen Time: ☐ Less than 2hrs ☐ 2-4hrs ☐ 5-7hrs ☐ 8-10hrs ☐ More than 10hrs

Purpose of Smart Phone Use: ☐ Social media ☐ Gaming ☐ Messaging/Calls ☐ Work/Study

Birth Order: ☐ First child ☐ Middle child ☐ Last child

Number of Siblings: _____

Age Gap with the Sibling: _____

Sibling's Information

Note: When asked about a sibling, you are required to keep ONE sibling in your mind with reference to whom you will be answering the questions related to phubbing behavior and sibling relationship.

Age of the sibling: _____

Gender of the sibling: ☐ Male ☐ Female

Birth Order of the sibling: ☐ First child ☐ Middle child ☐ Last child

Education Level of the sibling: ☐ Undergraduate ☐ Graduate ☐ Postgraduate

Marital Status of the sibling: ☐ Single ☐ Married

Sibling Relationship type: ☐ Biological ☐ Step Sibling ☐ Adopted Sibling

Average Screen Time: ☐ Less than 2hrs ☐ 2-4hrs ☐ 5-7hrs ☐ 8-10hrs ☐ More than 10hrs

Appendix F

Generic Scale of Being Phubbed

Statement	1	2	3	4	5	6	7
1. My sibling seems to check their phones for messages and social media updates.							
2. My sibling seems to be using their phones to go online.							
3. My sibling places their phones where they can see them.							
4. My sibling seems worried that they will miss something important if they do not check their phones.							
5. My sibling seems like they lose awareness of their surroundings because of their phone use.							
6. My sibling seems like they have a difficult time putting their phones down.							
7. My sibling seems like they cannot stand leaving their phones alone.							
8. My sibling seems like they are “in their own worlds” using their phones.							
9. My sibling seems anxious if their phones are not nearby.							
10. My sibling pays attention to their phones rather than talking to me.							
11. My sibling would rather pay attention to their phones than talk to me.							
12. My sibling seems like they get rid of boredom by paying attention to their phones instead of me.							
13. My sibling seems like they feel content when they are paying attention to their phones instead of me.							
14. My sibling pays attention to their phones rather than focusing on me.							
15. My sibling seems like they get rid of stress by paying attention to their phones instead of me.							
16. My sibling seems like they feel good when they stop focusing on me and pay attention to their phones instead.							
17. My sibling shifts their attention from me to their phones.							
18. I tell my sibling that they interact with their phones too much.							
19. I have conflicts with my sibling because they are using their phones.							
20. I find myself thinking “I’ve had enough” when my sibling is using their phones.							
21. My sibling use their phone even though they know it irritates me							
22. My sibling seem like they get irritated if I ask them to get off their phones and talk to me							

Appendix G

Lifespan Sibling Relationship Scale

Statements	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
1. My sibling makes me happy.					
2. My sibling's feelings are very important to me.					
3. I enjoy my relationship with my sibling.					
4. I am proud of my sibling.					
5. My sibling and I have a lot of fun together.					
6. My sibling frequently makes me very angry.					
7. I admire my sibling.					
8. I like to spend time with my sibling.					
9. I presently spend a lot of time with my sibling.					
10. I call my sibling on the telephone frequently.					
11. My sibling and I share secrets.					
12. My sibling and I do a lot of things together.					
13. I never talk about my problems with my sibling.					
14. My sibling and I borrow things from each other.					
15. My sibling and I 'hang out' together.					
16. My sibling talks to me about personal problems.					
17. My sibling is a good friend.					
18. My sibling is very important in my life.					
19. My sibling and I are not very close.					
20. My sibling is one of my best friends.					
21. My sibling and I have a lot in common.					
22. I believe I am very important to my sibling.					
23. I know that I am one of my sibling's best friends.					
24. My sibling is proud of me.					

Appendix H

UCLA Loneliness Scale – Version 3

Note: Respond to the following questions with respect to your sibling interactions.

Statements	Never	Rarely	Sometimes	Often
1. How often do you feel you are in tune with the people around me?	1	2	3	4
2. How often do you feel that you lack companionship?	1	2	3	4
3. How often do you feel there is no one you can turn to?	1	2	3	4
4. How often do you feel alone?	1	2	3	4
5. How often do you feel part of a group of friends?	1	2	3	4
6. How often do you feel that you have a lot in common with the people around you?	1	2	3	4
7. How often do you feel you are no longer close to anyone?	1	2	3	4
8. How often do you feel your interests and ideas are not shared by those around you?	1	2	3	4
9. How often do you feel outgoing and friendly?	1	2	3	4
10. How often do you feel close to people?	1	2	3	4
11. How often do you feel left out?	1	2	3	4
12. How often do you feel that your relationships with others are not meaningful?	1	2	3	4
13. How often do you feel no one know you well?	1	2	3	4
14. How often do you feel isolated from others?	1	2	3	4
15. How often do you feel that you can find companionship when you want it?	1	2	3	4
16. How often do you feel that there are people who really understand you?	1	2	3	4
17. How often do you feel shy?	1	2	3	4
18. How often do you feel that people are around you but not with you?	1	2	3	4
19. How often do you feel there are people you can talk to?	1	2	3	4
20. How often do you feel there are people you can turn to?	1	2	3	4